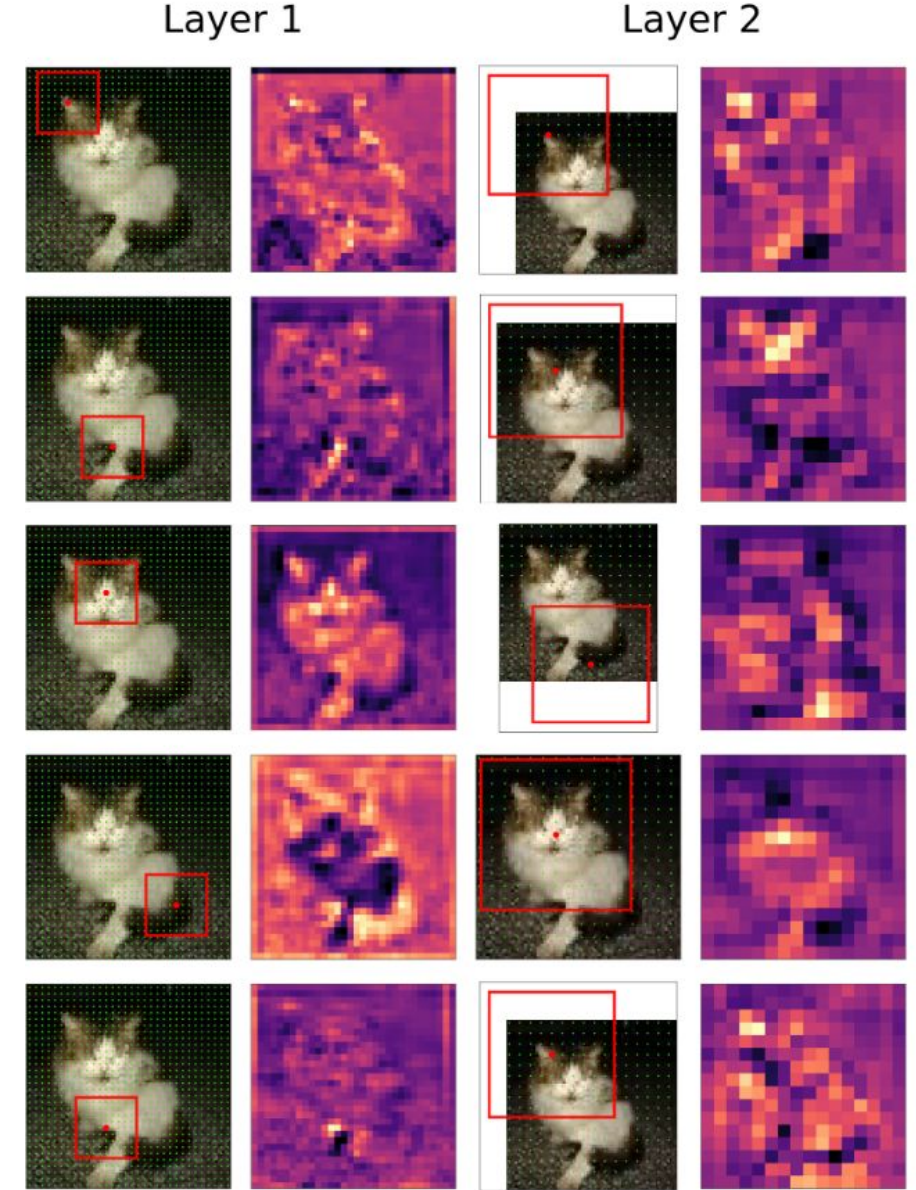
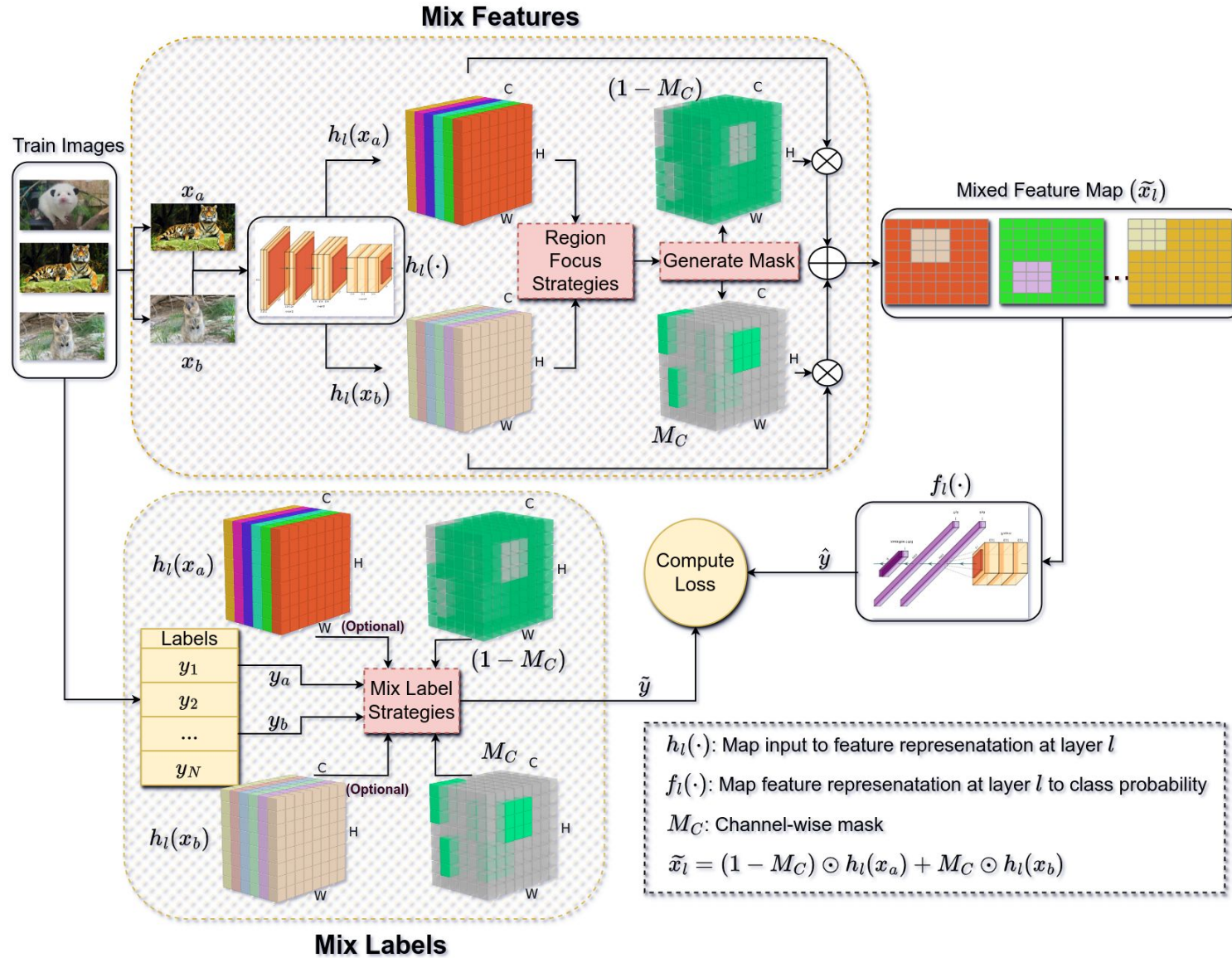


FMMix: A Novel Augmentation Strategy for Enhancing Feature Diversity to Boost Model Generalization for Image Data

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Outline

1. Introduction
2. Method Description
3. Experimental Setup
4. Experimental Results
5. Conclusion

Introduction

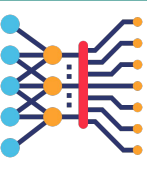
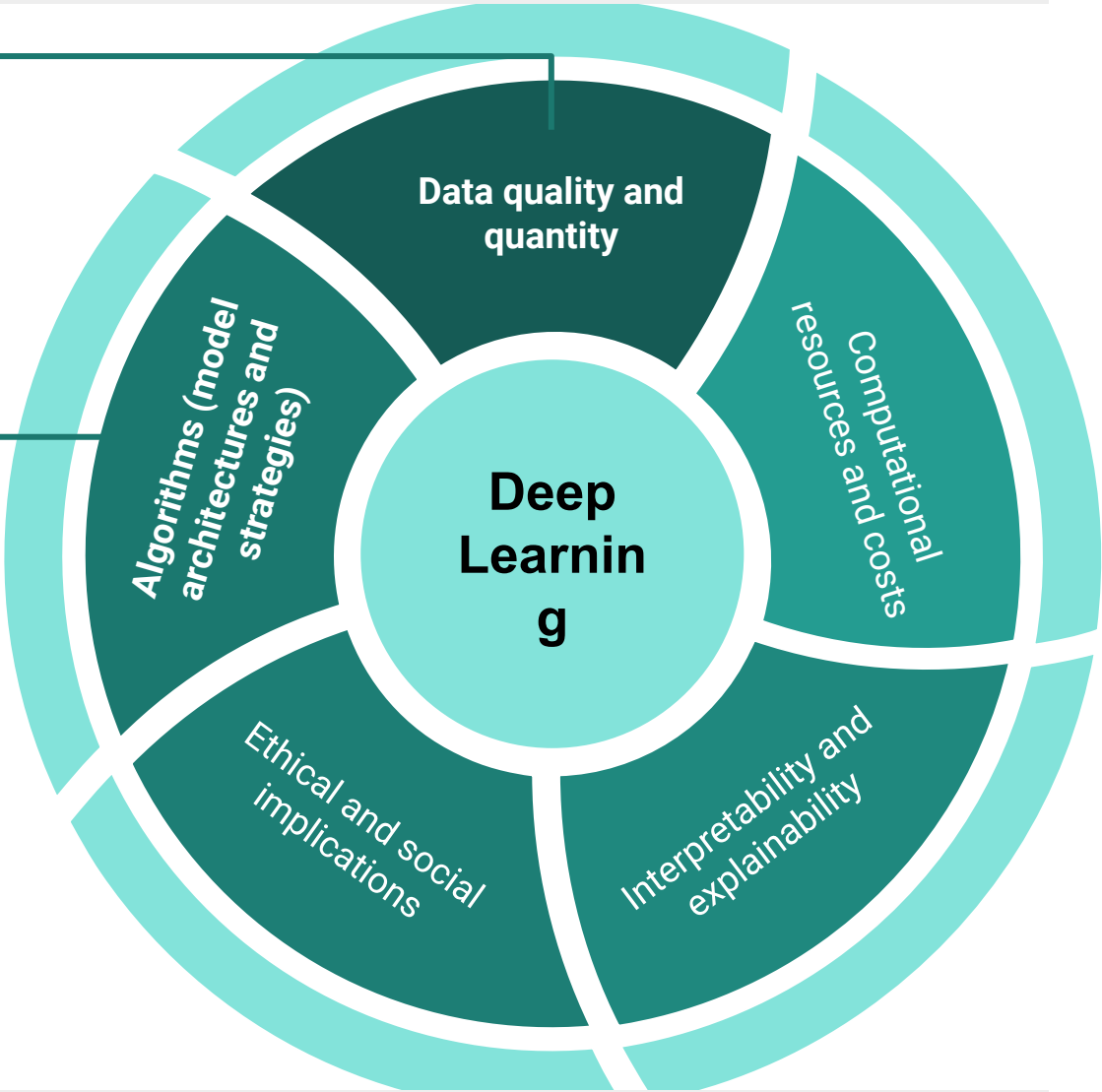


Data Quality & Quantity

- Essential for deep learning success.
- Insufficient data leads to poor generalization and overfitting
- Low-quality data results in learning from noise and biases, affecting accuracy

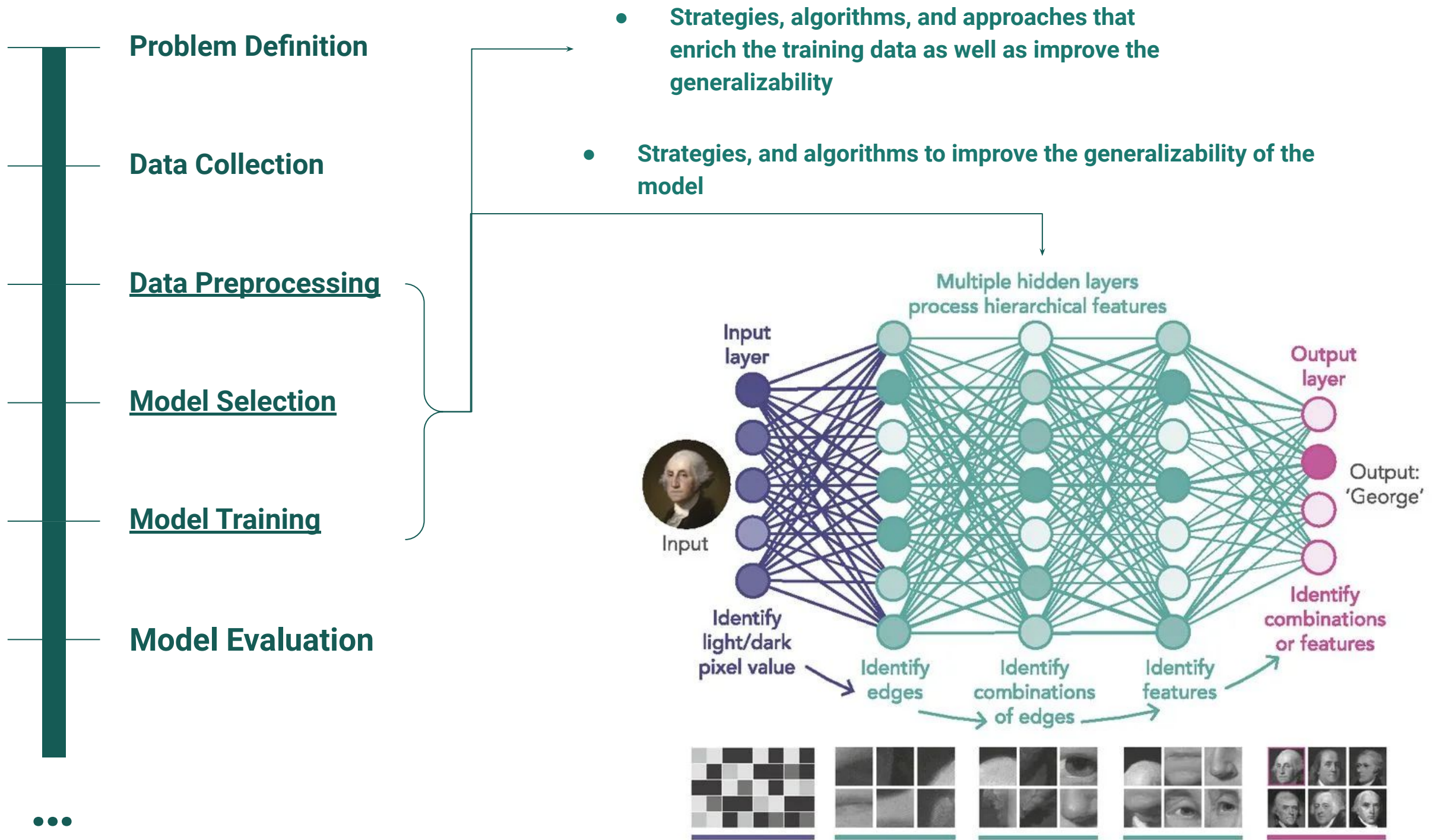
- Focus on improving the generalization
- Goal is to maximize the performance using the same dataset

Research Direction



Algorithms (model architectures and strategies)

- More complex models, with many parameters, are more prone to overfitting
- Adapting models to new tasks or datasets while retaining learned knowledge without extensive retraining can be difficult



Data augmentation

Illumination



Deformation



Occlusion



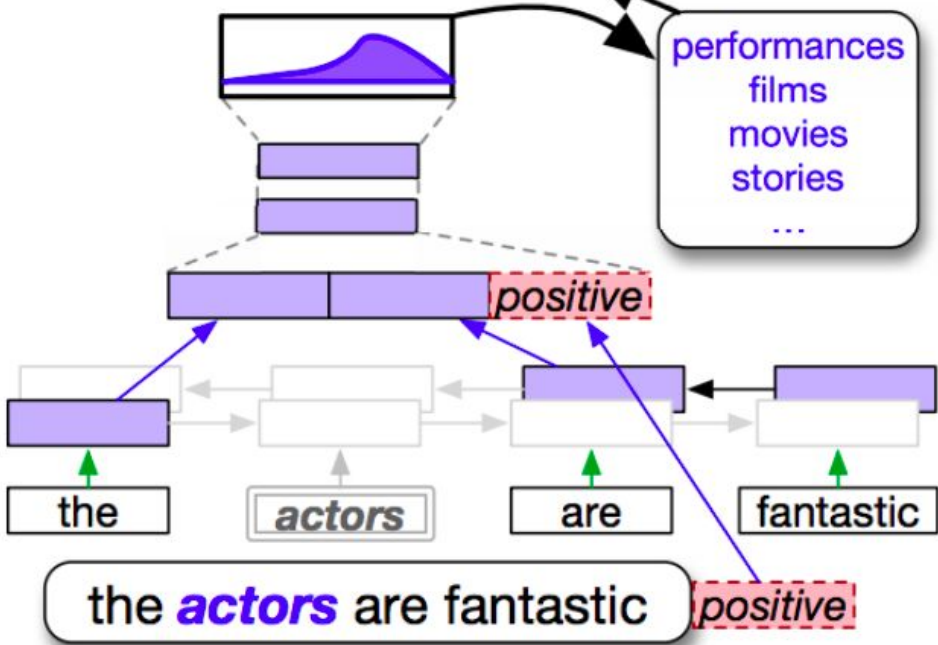
Background



the **performances** are fantastic
the **films** are fantastic
the **movies** are fantastic
the **stories** are fantastic
...

positive

performances
films
movies
stories
...



Flipping

Rotating

Cropping

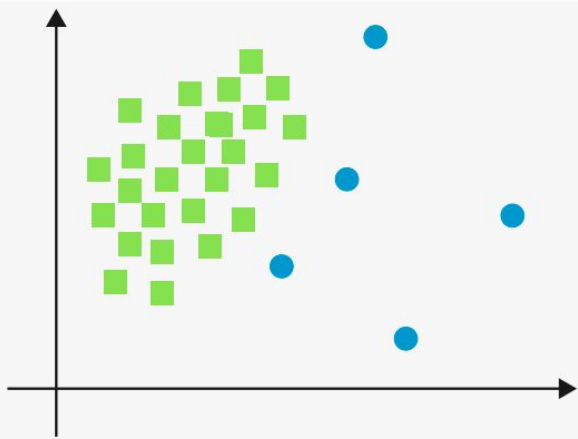


Original

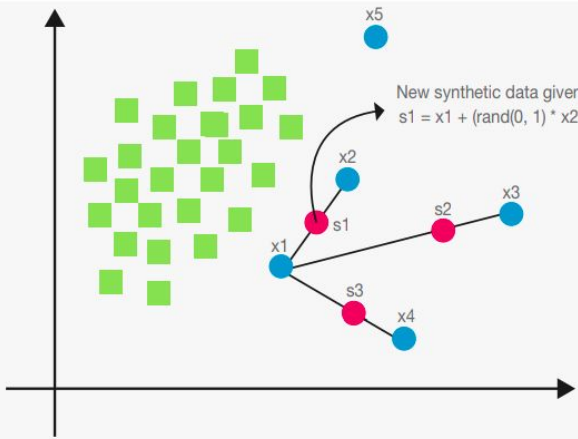
Color distortion

Sharpening

Random erasing

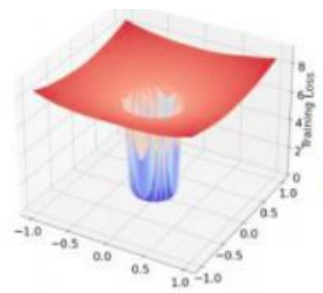
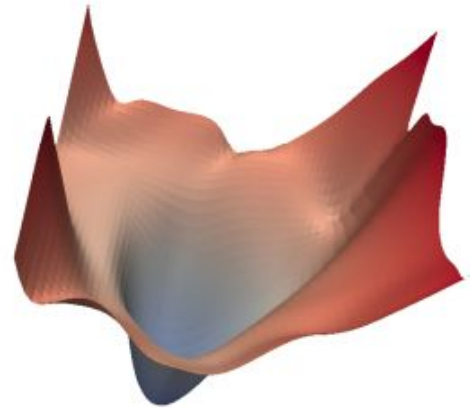
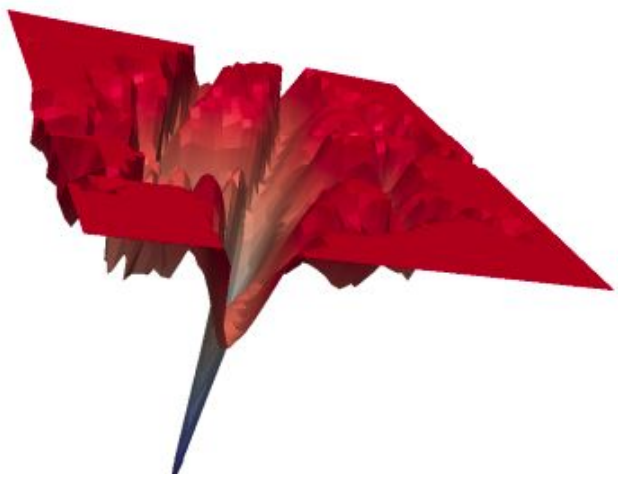
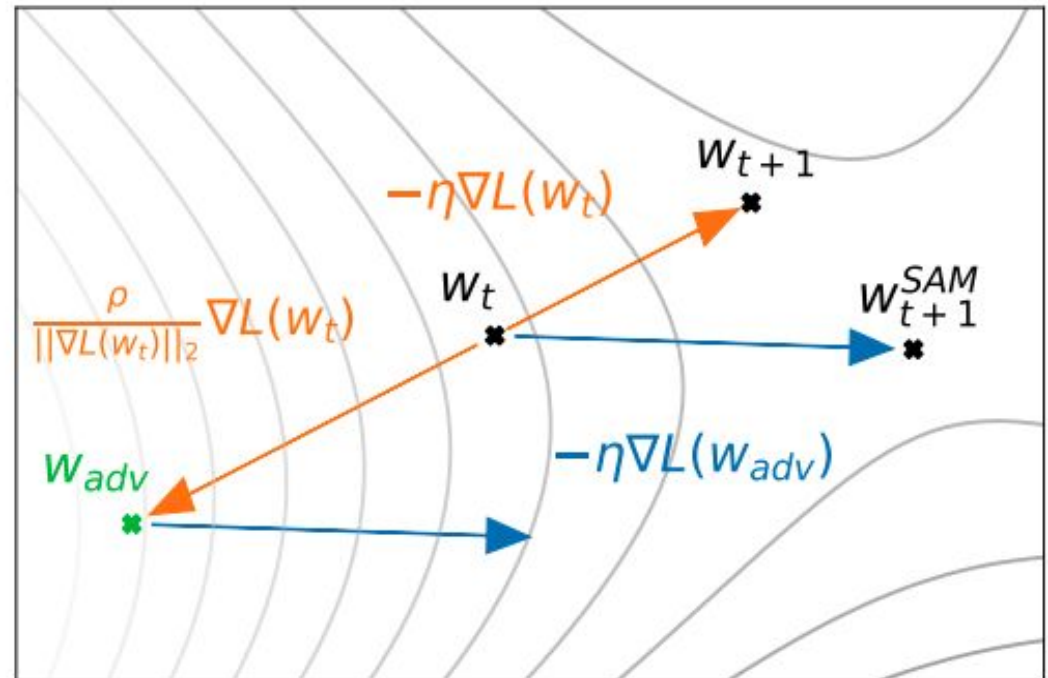
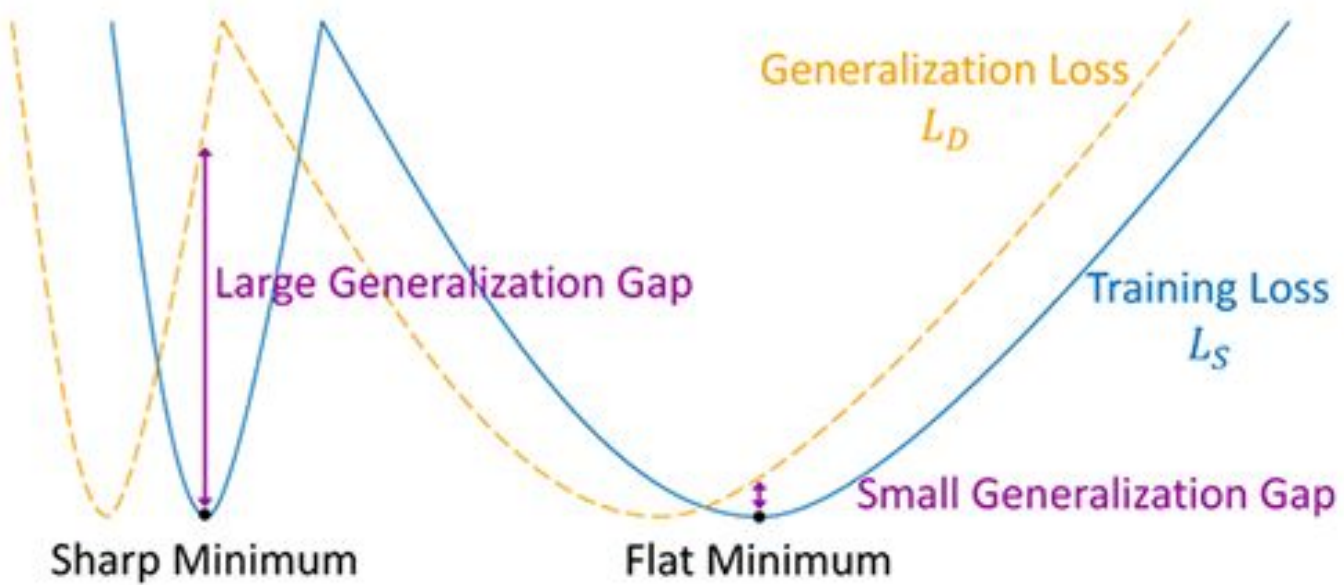


a) Class Imbalance

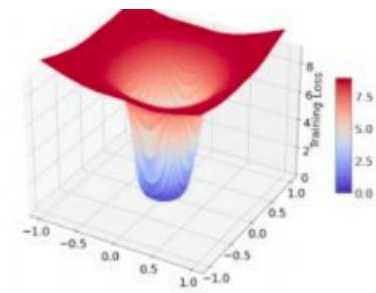


b) SMOTE

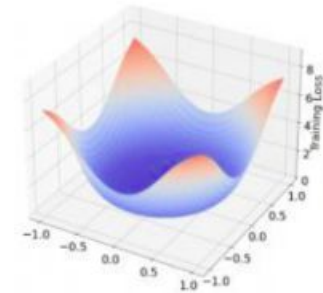
Optimizer



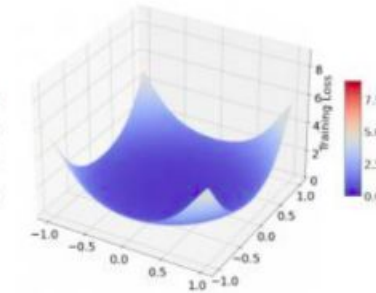
(b) ViT



(c) Mixer



(d) ViT-SAM

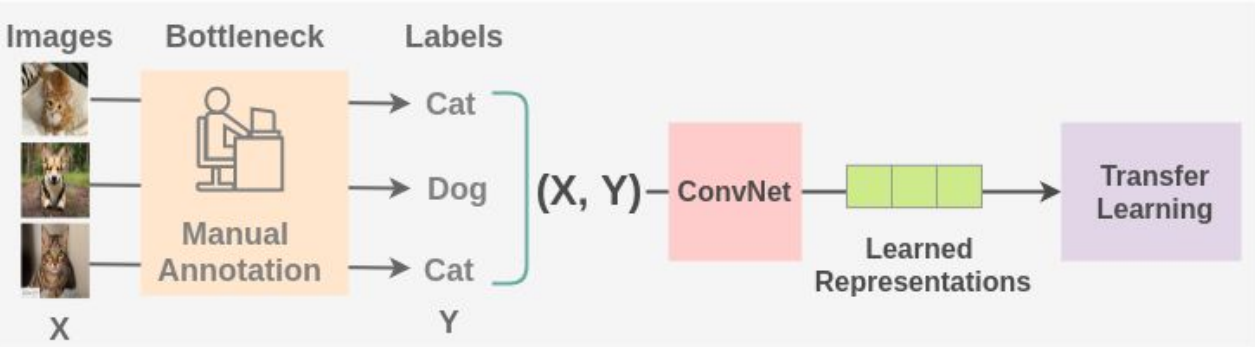


(e) Mixer-SAM

Foret, P., Kleiner, A., Mobahi, H., & Neyshabur, B. (2020). Sharpness-Aware Minimization for Efficiently Improving Generalization. *ArXiv, abs/2010.01412*.

Self-supervise

Supervised Learning Workflow



Self-Supervised Learning Workflow

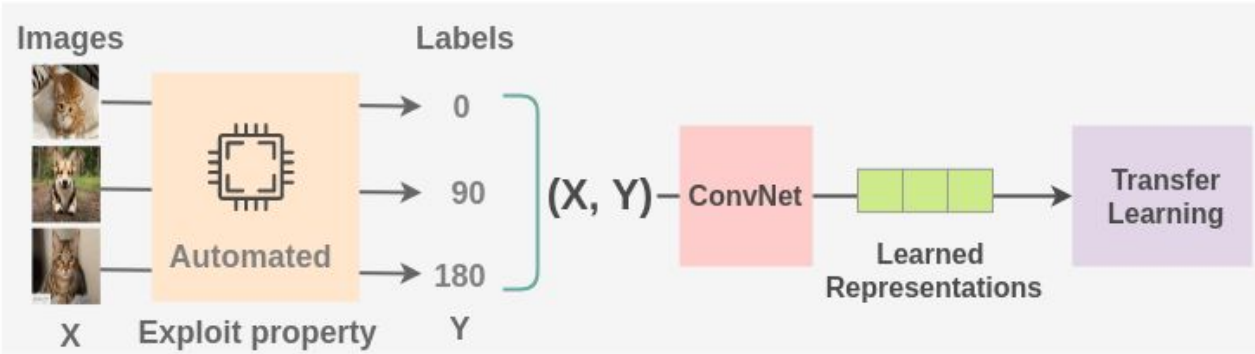
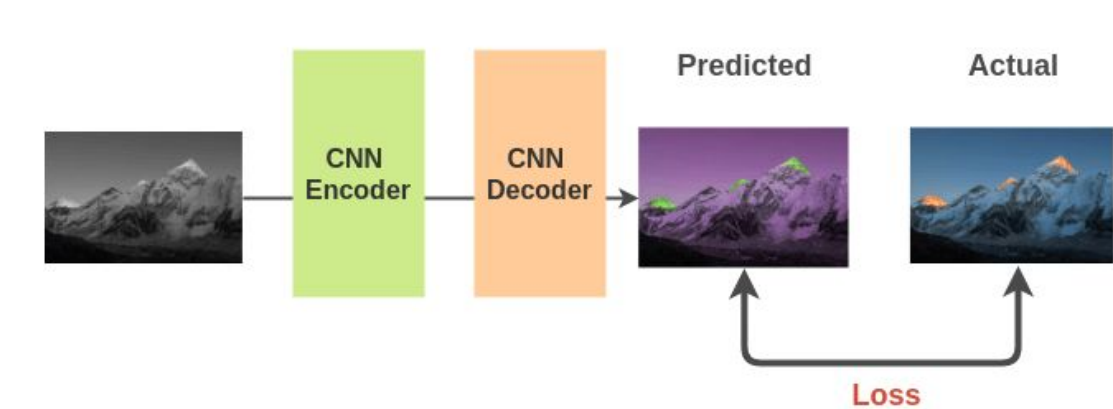
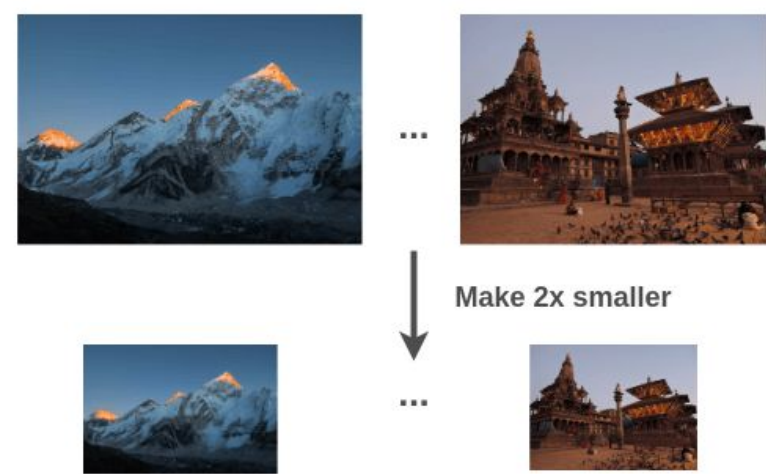


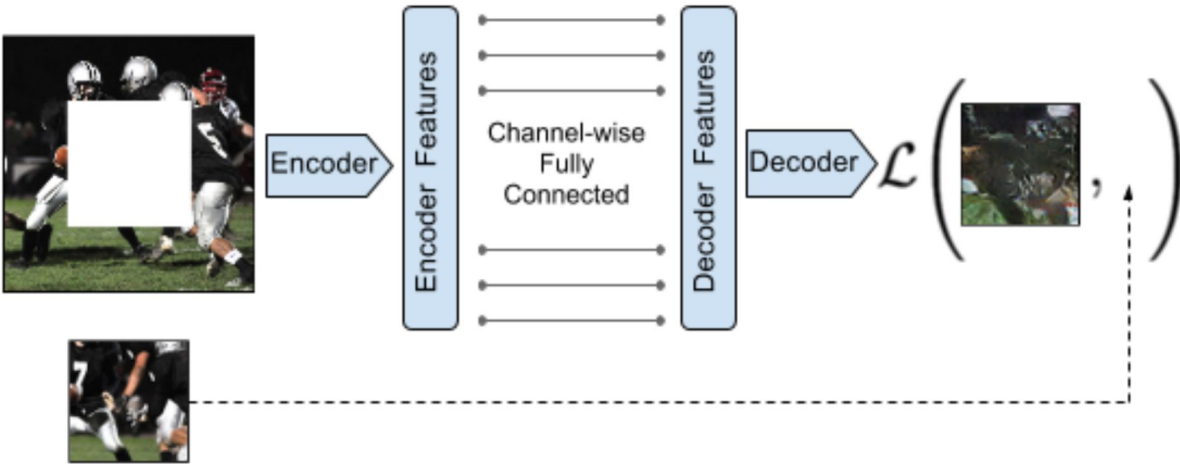
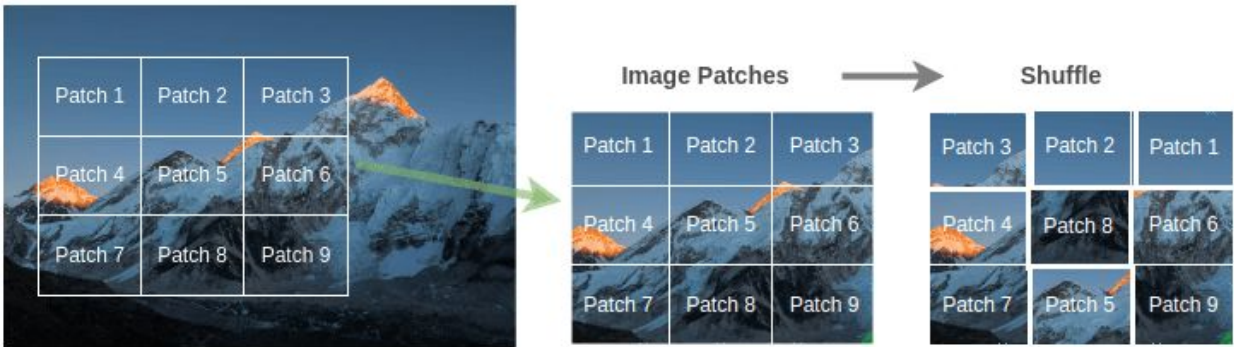
Image Colorization



Training Data Generation for Superresolution



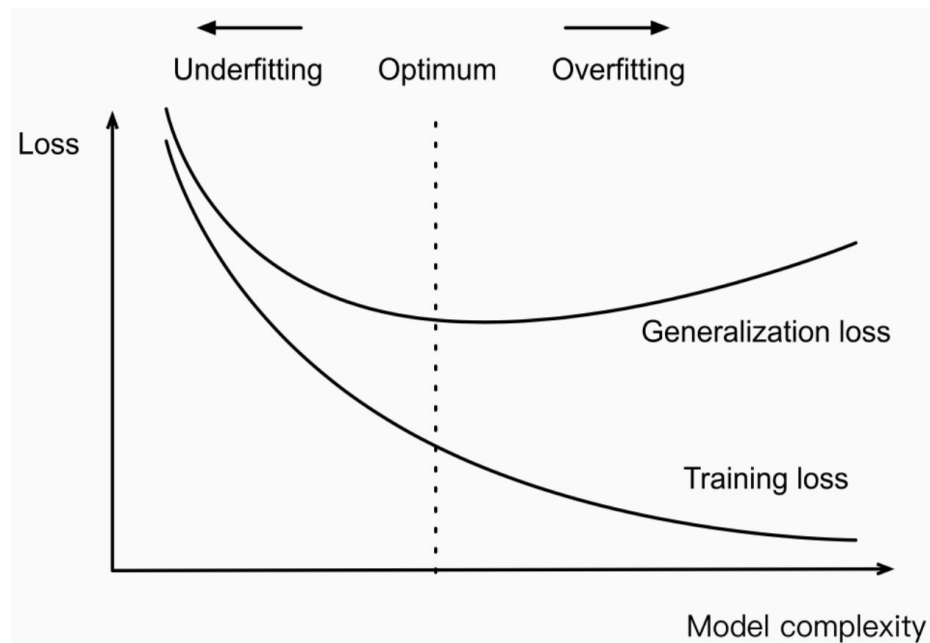
Jigsaw Data Generation



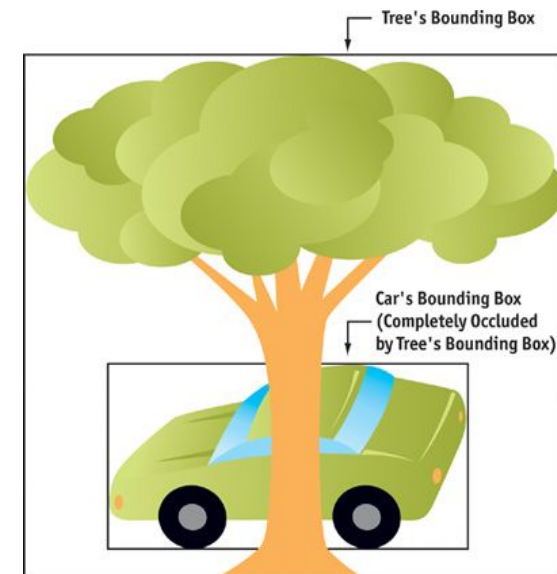
Overfitting

Poor generalization to unseen data

- Insufficient Training Data
- Noisy Data
- Lack of Regularization
- Object Occlusion



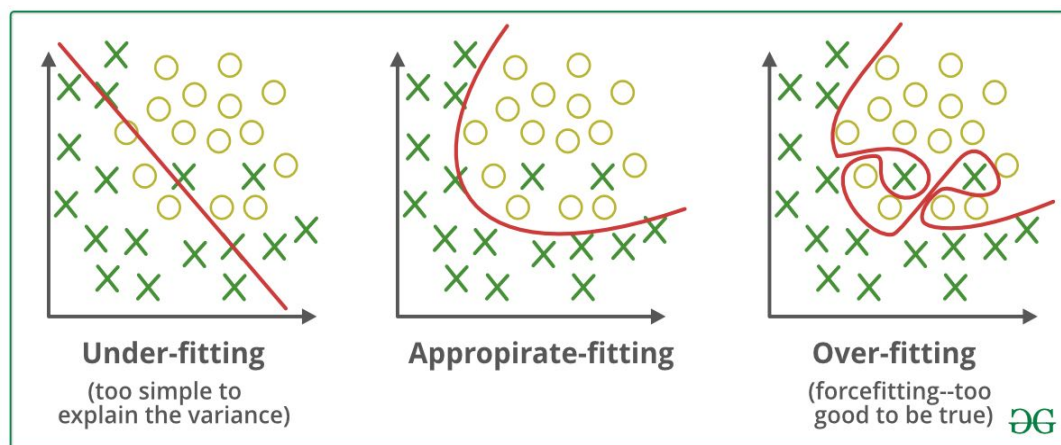
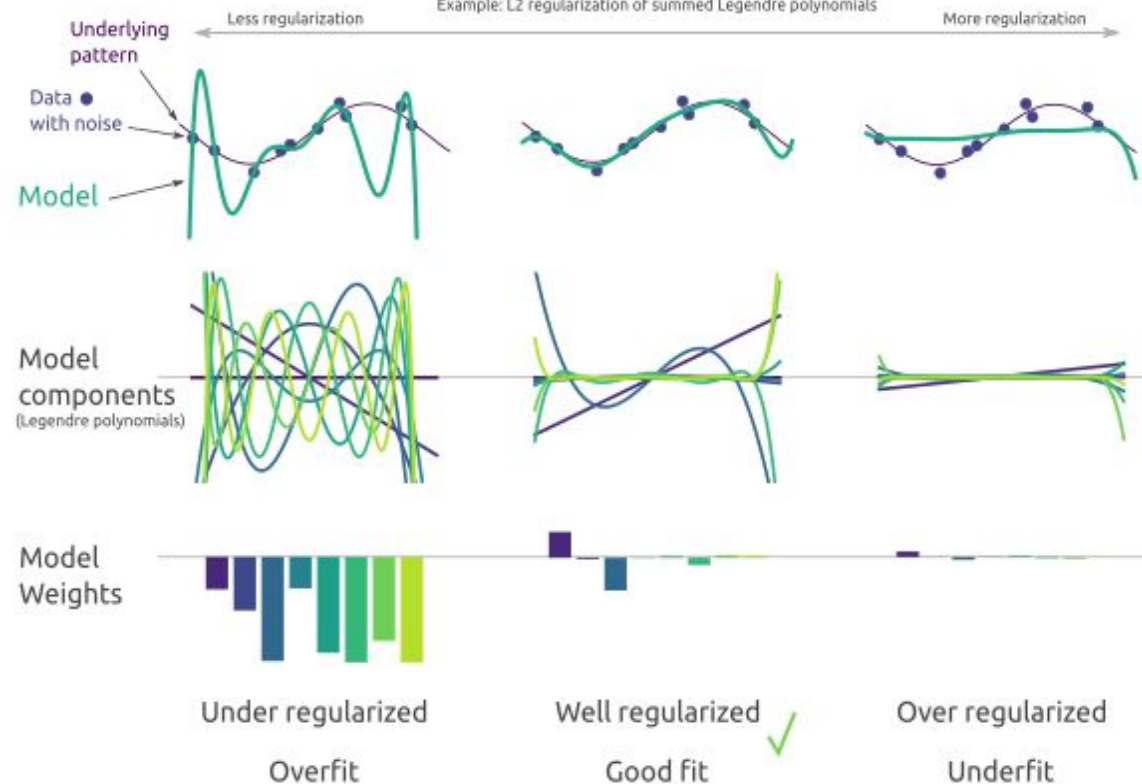
Noisy Data



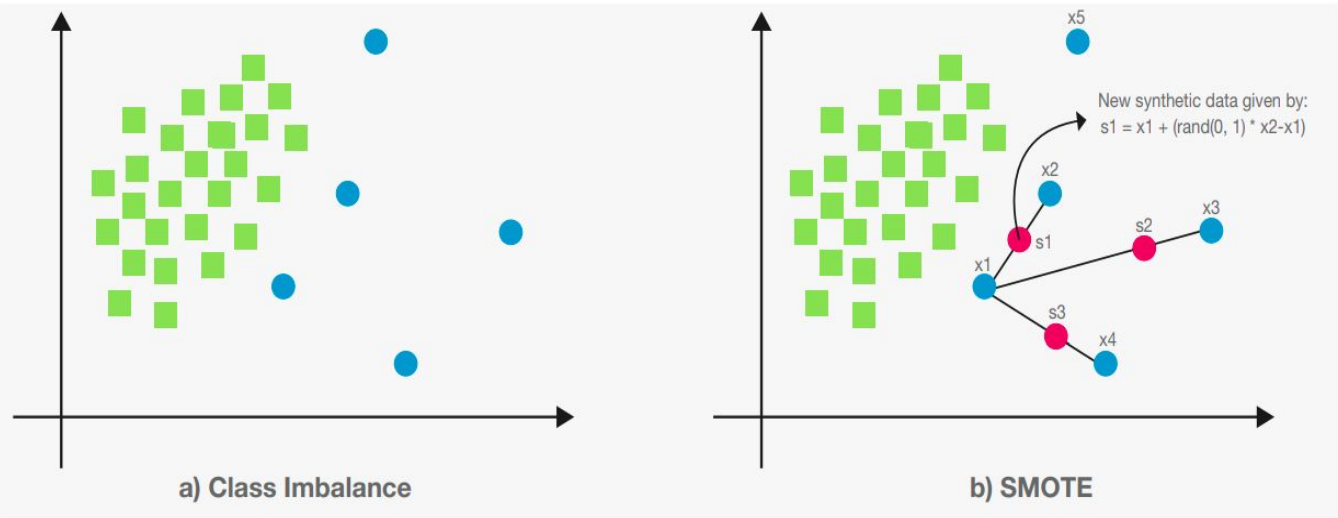
Object Occlusion

Model Regularization

Example: L2 regularization of summed Legendre polynomials

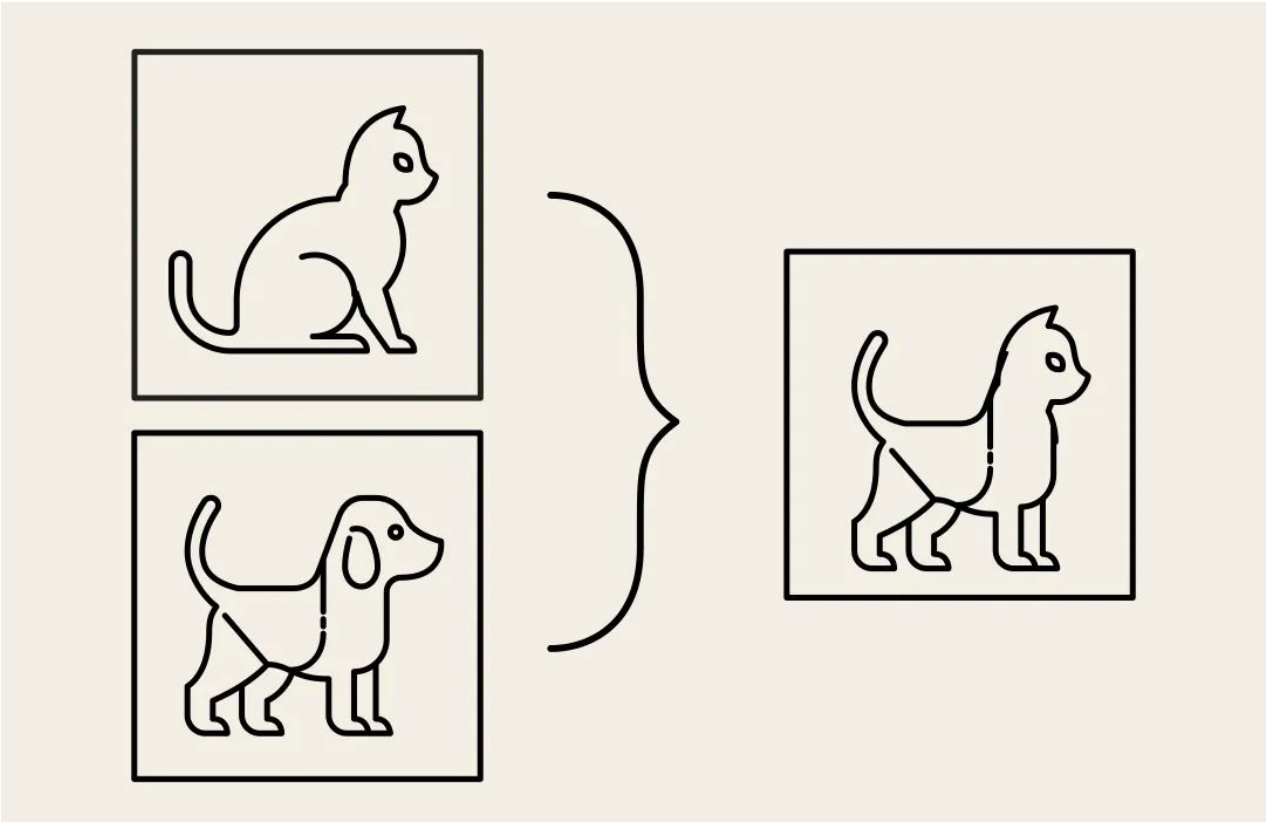
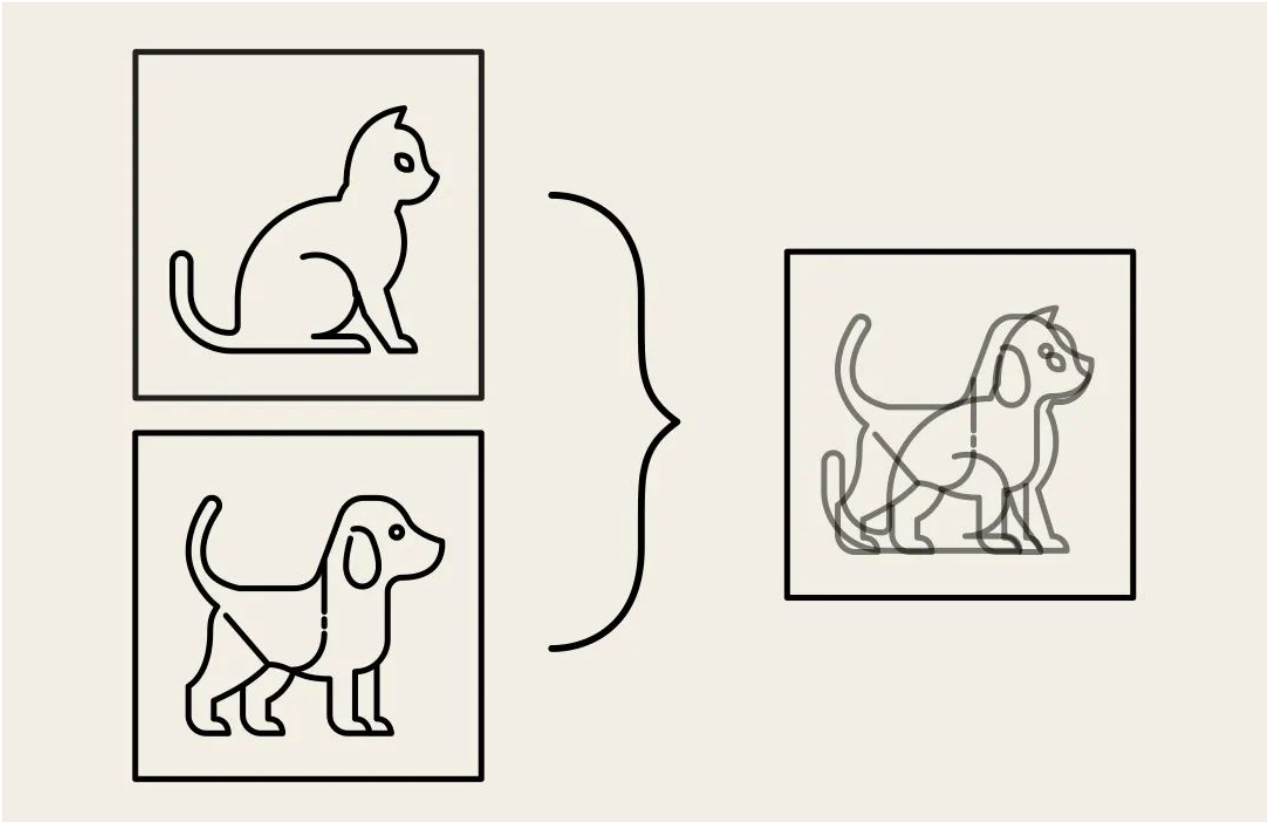


Augmentation

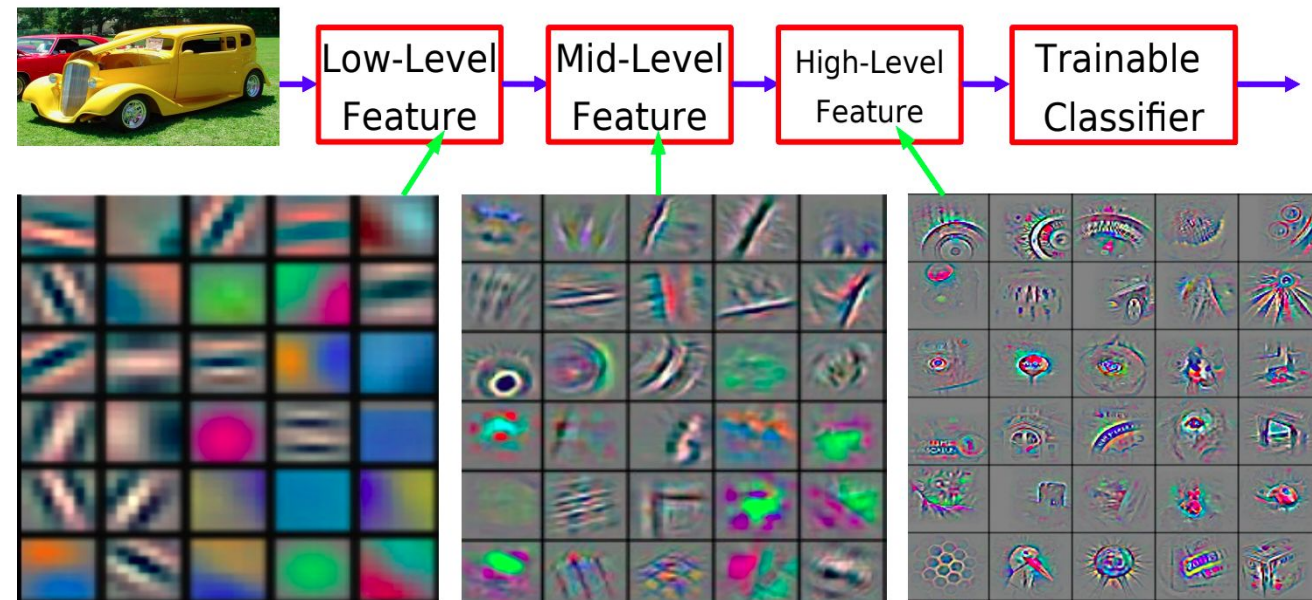
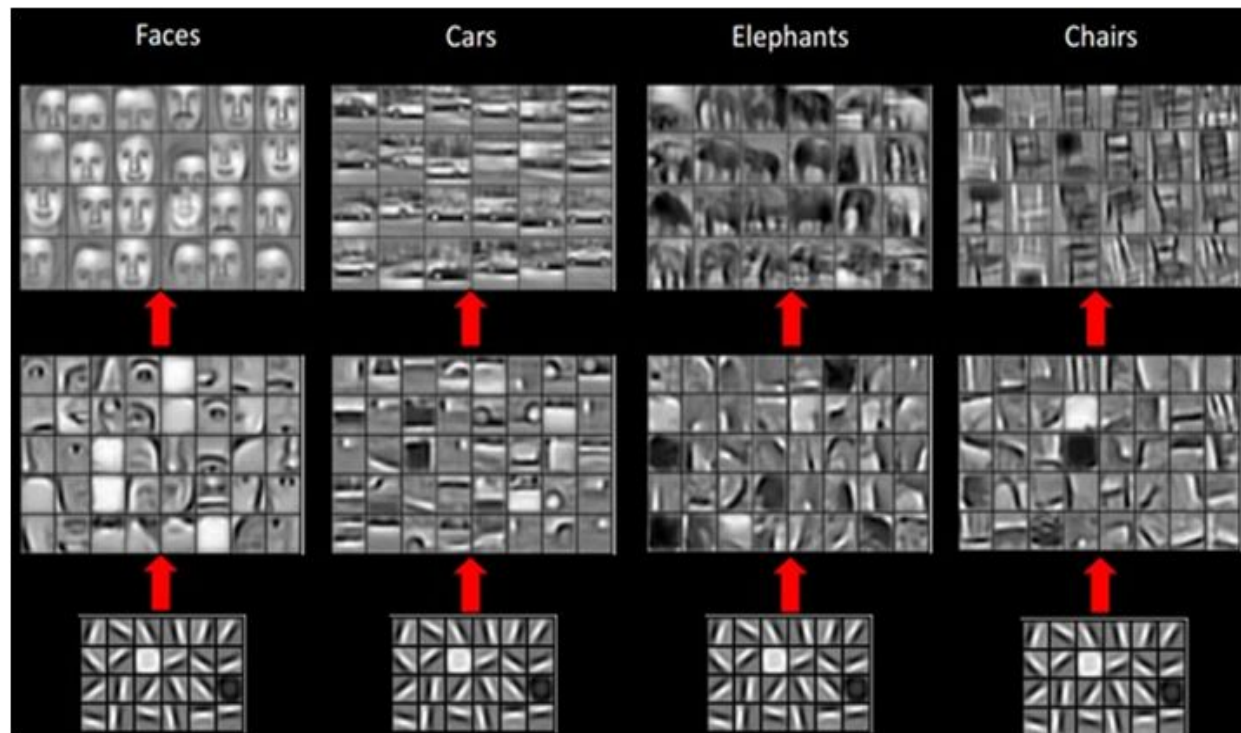
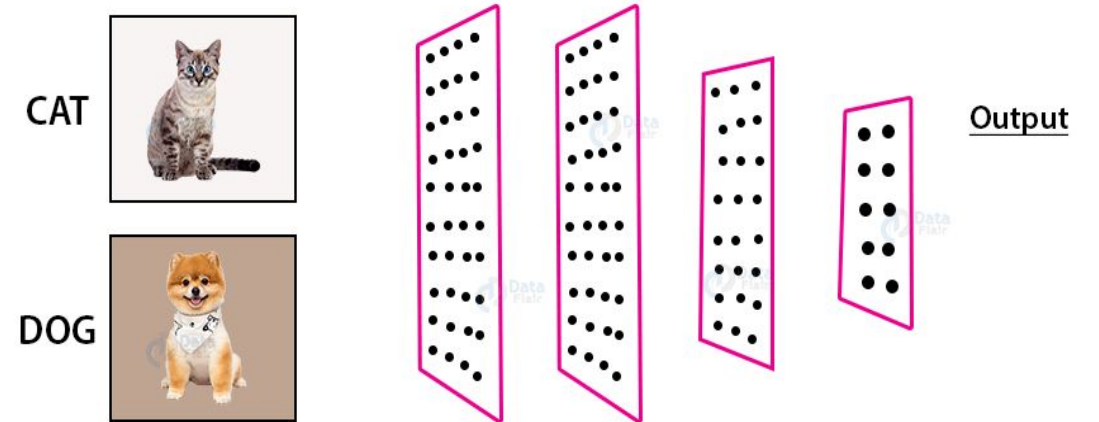
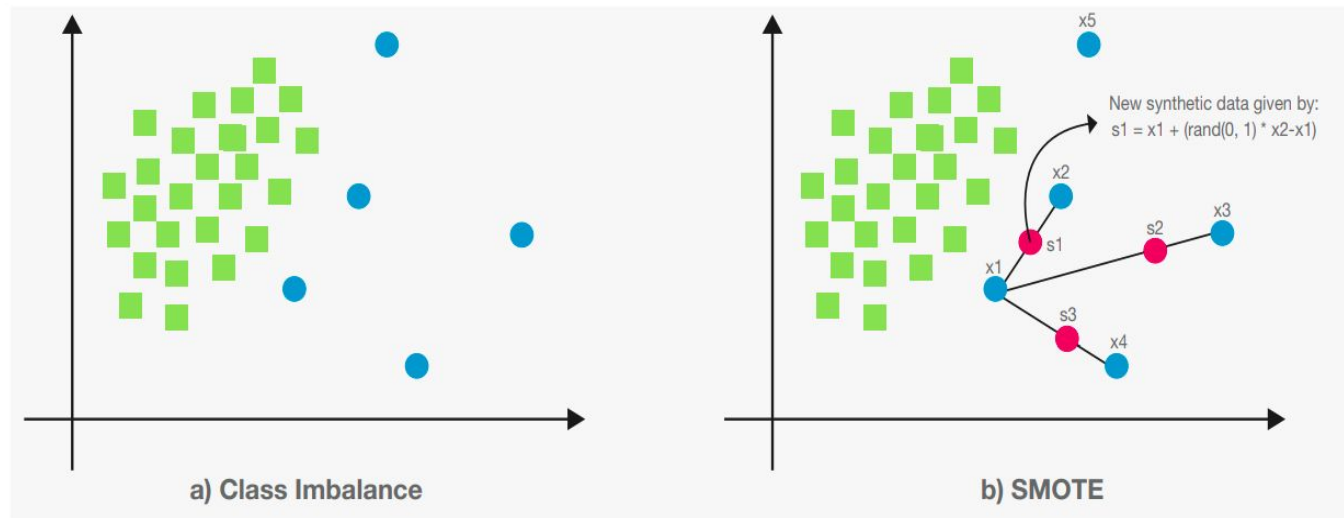


Mixup

CutMix

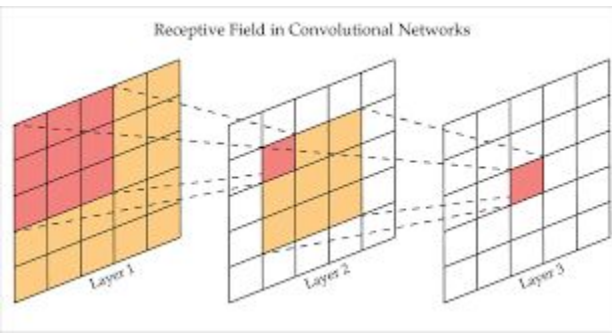


Augmentation



CNN receptive field

How does feature-level augmentation affect deep learning models compared to input-level techniques, and what can we learn about the effectiveness of utilizing higher-level information in augmentation?



Layer 1

Layer 2

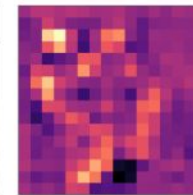
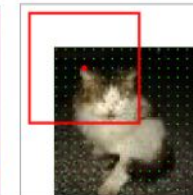
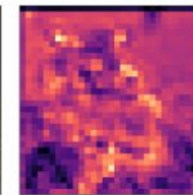
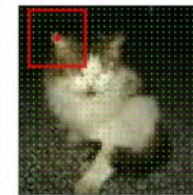
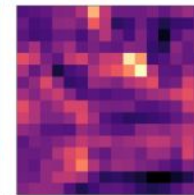
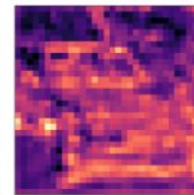
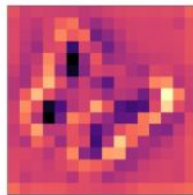
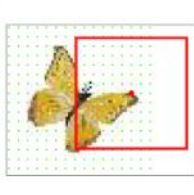
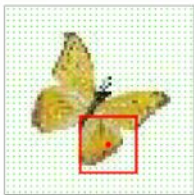
Layer 1

Layer 2

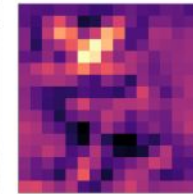
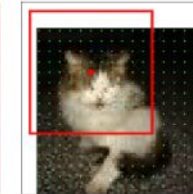
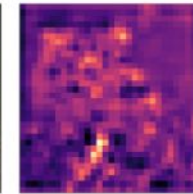
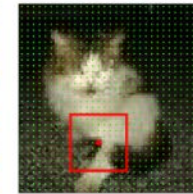
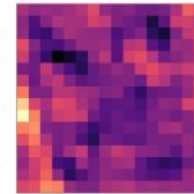
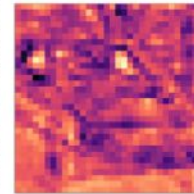
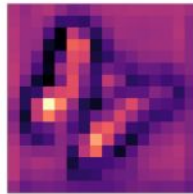
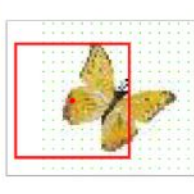
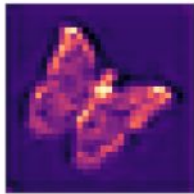
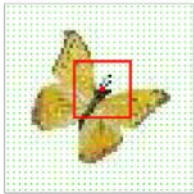
Layer 1

Layer 2

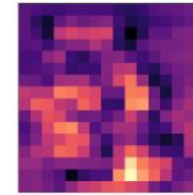
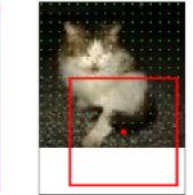
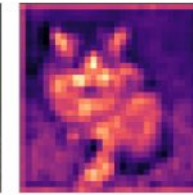
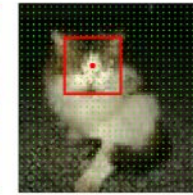
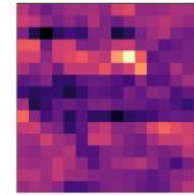
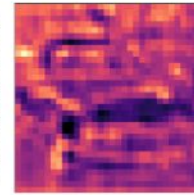
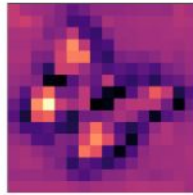
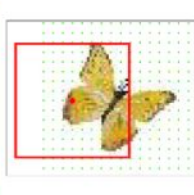
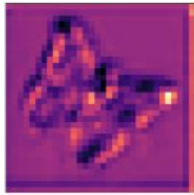
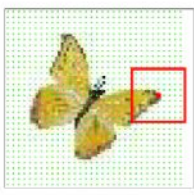
Top 1



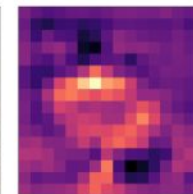
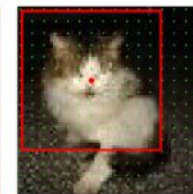
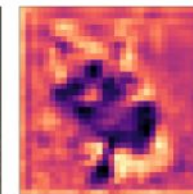
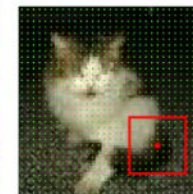
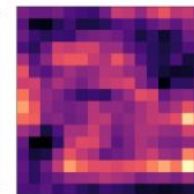
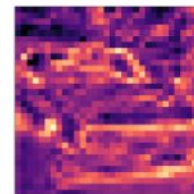
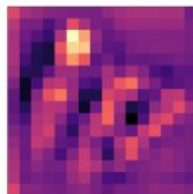
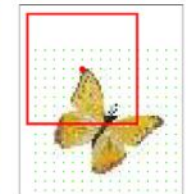
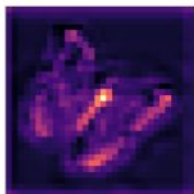
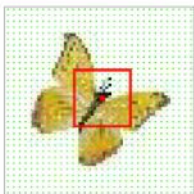
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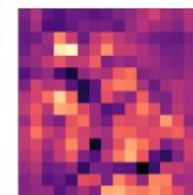
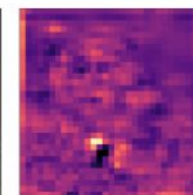
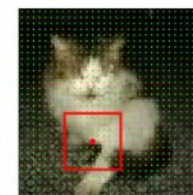
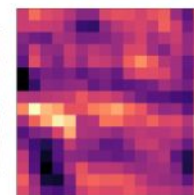
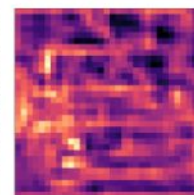
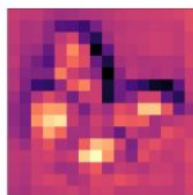
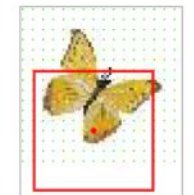
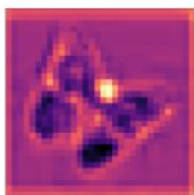
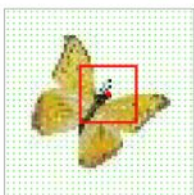
Top 3



Top 4



Top 5



Feature-space augmentation

How does feature-level augmentation affect deep learning models compared to input-level techniques, and what can we learn about the effectiveness of utilizing higher-level information in augmentation?

Layer Specific Augmentation?

TABLE 3. Accuracy (%) on CIFAR100 using Mixup and CutMix at various layers to evaluate their impact on model generalization. (Baseline accuracy: 74.57%; * denotes input level augmentation.)

Layer	Mixup	CutMix
[0]*	77.48 (+2.91)	78.17 (+3.60)
[1]	77.99 (+3.42)	79.20 (+4.63)
[2]	78.10 (+3.53)	78.17 (+3.60)
[3]	76.20 (+1.63)	76.84 (+2.27)
[4]	77.11 (+2.54)	77.16 (+2.59)

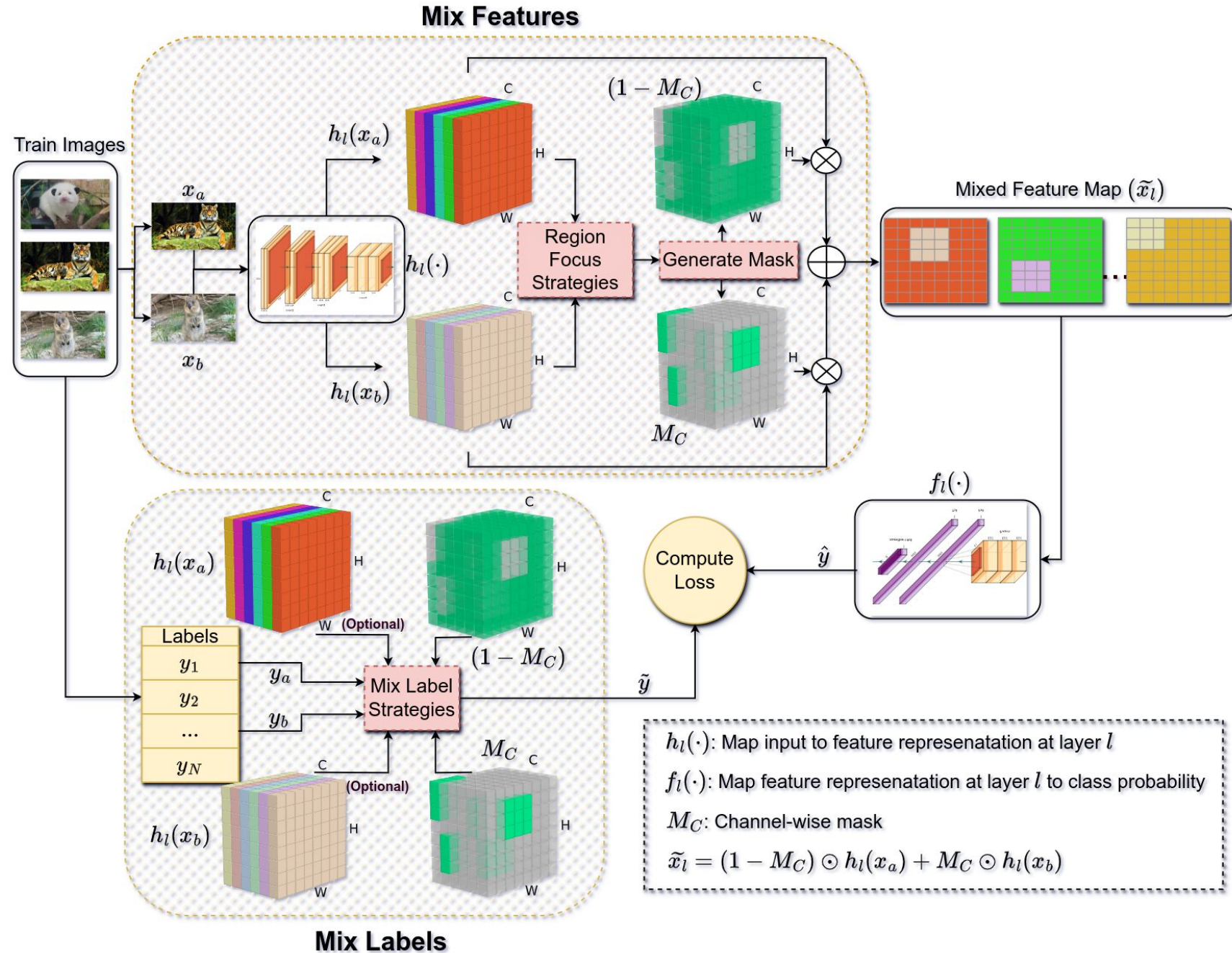
Combination Augmentation?

TABLE 4. Accuracy (%) on CIFAR100 using Mixup and CutMix across various combinations of layers to assess their influence on model generalization. (Baseline accuracy: 74.57%.)

Layer	Mixup	CutMix
[1, 2]	78.45 (+3.88)	79.02 (+4.45)
[2, 3]	78.35 (+3.78)	78.10 (+3.53)
[3, 4]	77.90 (+3.33)	77.60 (+3.03)
[1, 2, 3]	76.11 (+1.54)	79.19 (+4.62)
[2, 3, 4]	Fail	78.38 (+3.81)
[1, 2, 3, 4]	79.00 (+4.43)	79.40 (+4.83)

Method Description

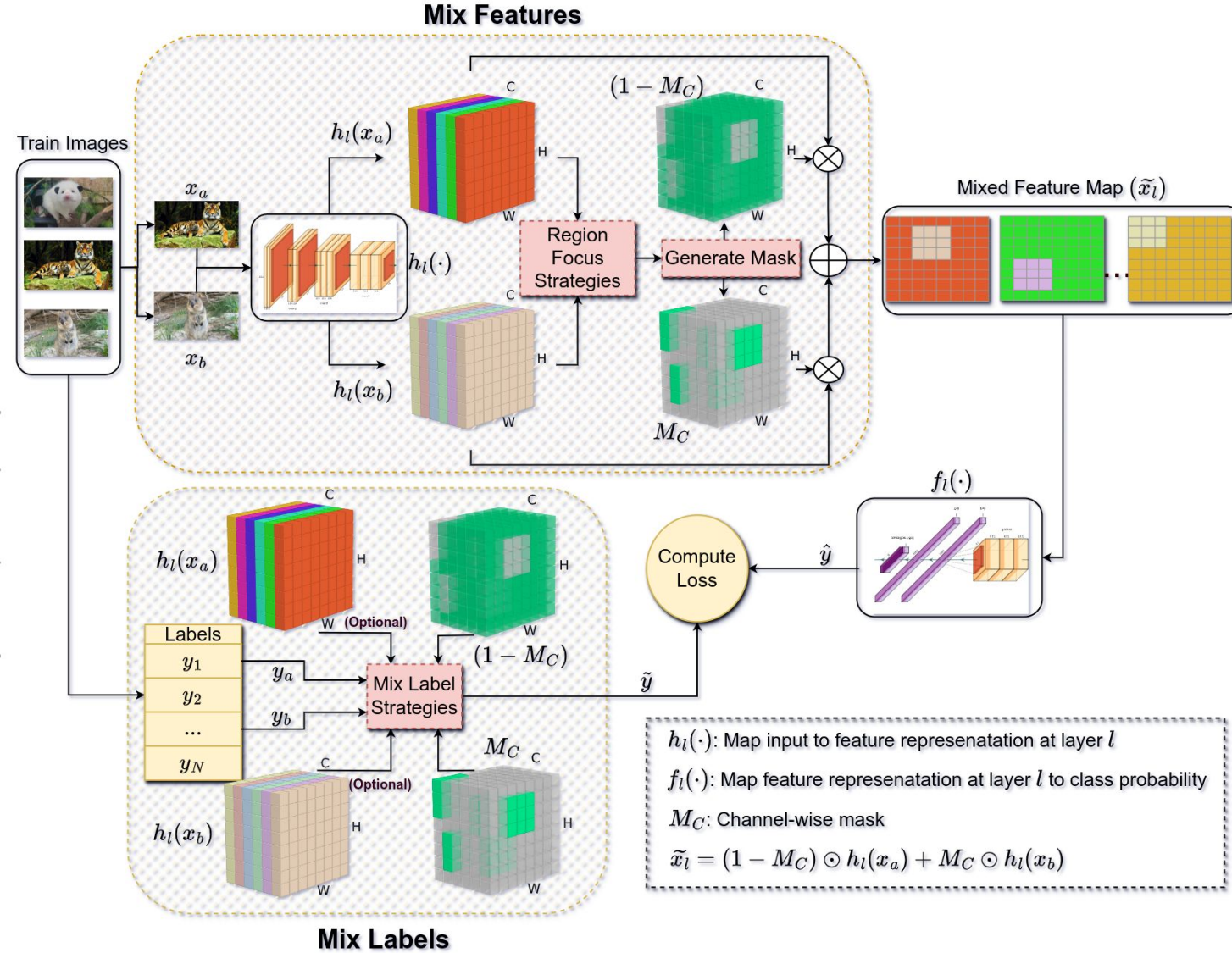
Overview of the FMMix augmentation process



Overview of the FMMix augmentation process

TABLE 1. Comparison of four FMMix variants based on our proposed feature map and label mixing methods.

	FMMix1	FMMix2	FMMix3	FMMix4
Max position	✓	✓	✗	✗
Top k patches	✗	✗	✓	✓
Feature map area ratio	✓	✗	✓	✗
Semantic information	✗	✓	✗	✓



Redefine the problem

$$(\tilde{x}, \tilde{y}) = \tilde{g}((x_a, y_a), (x_b, y_b)),$$

$$\tilde{y} = p_a y_a + p_b y_b,$$

$$\tilde{x} = \lambda x_a + (1 - \lambda) x_b,$$

$$p_a = \lambda, \quad p_b = 1 - \lambda.$$

$$\tilde{x} = \mathbf{M} \odot x_a + (1 - \mathbf{M}) \odot x_b,$$

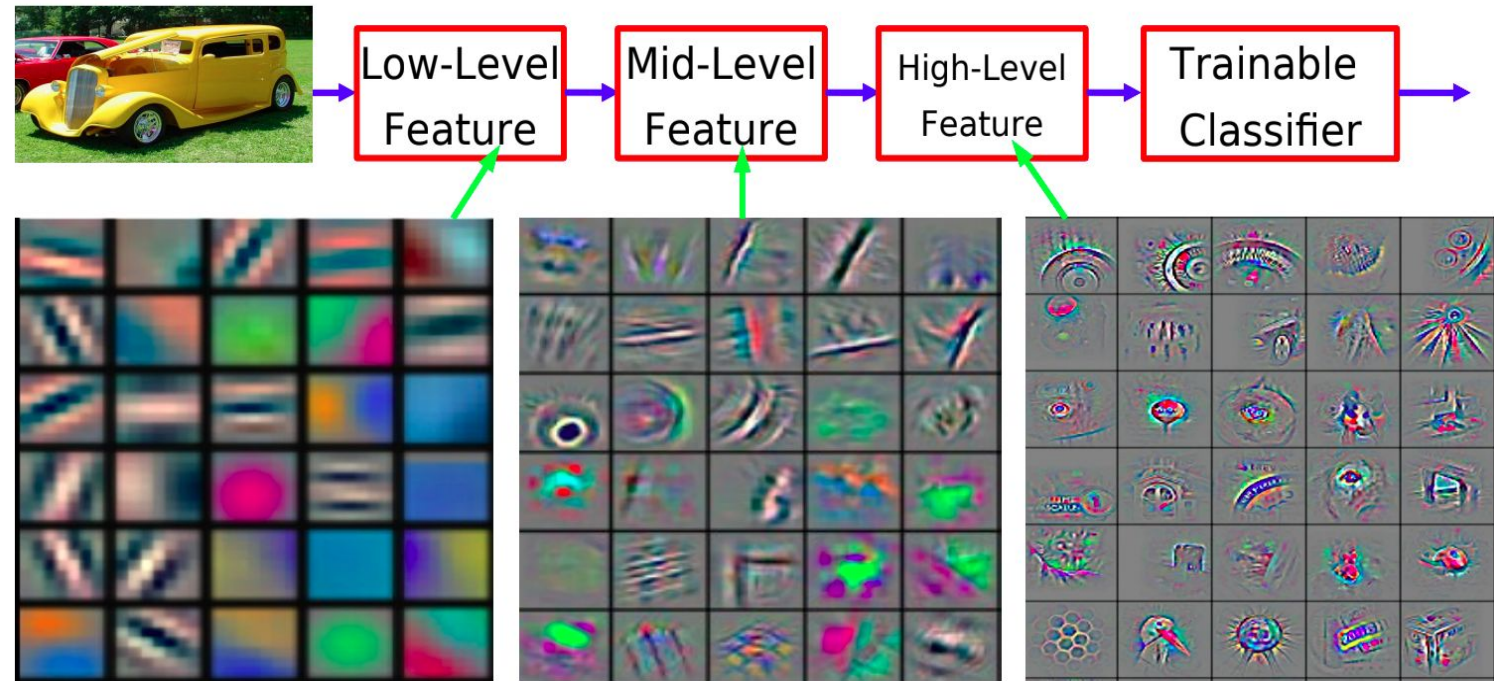
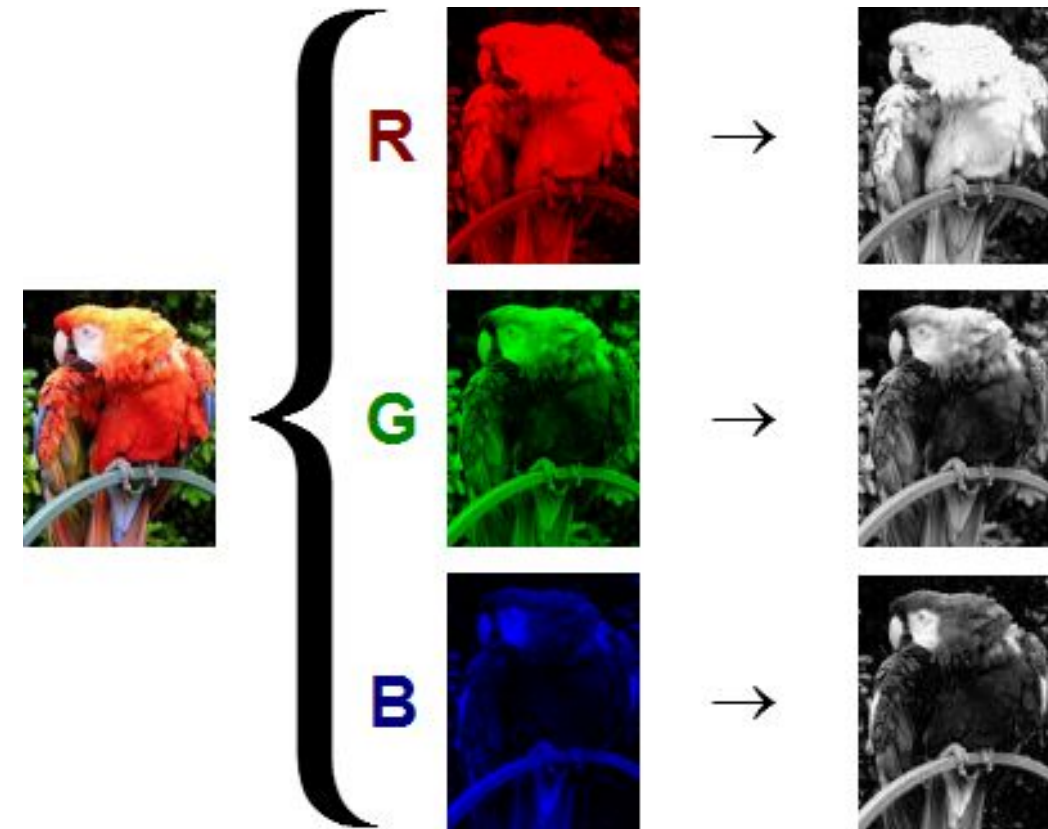
$$p_a = \lambda, \quad p_b = 1 - \lambda,$$

$$\hat{y} = f_l(h_l(x)),$$

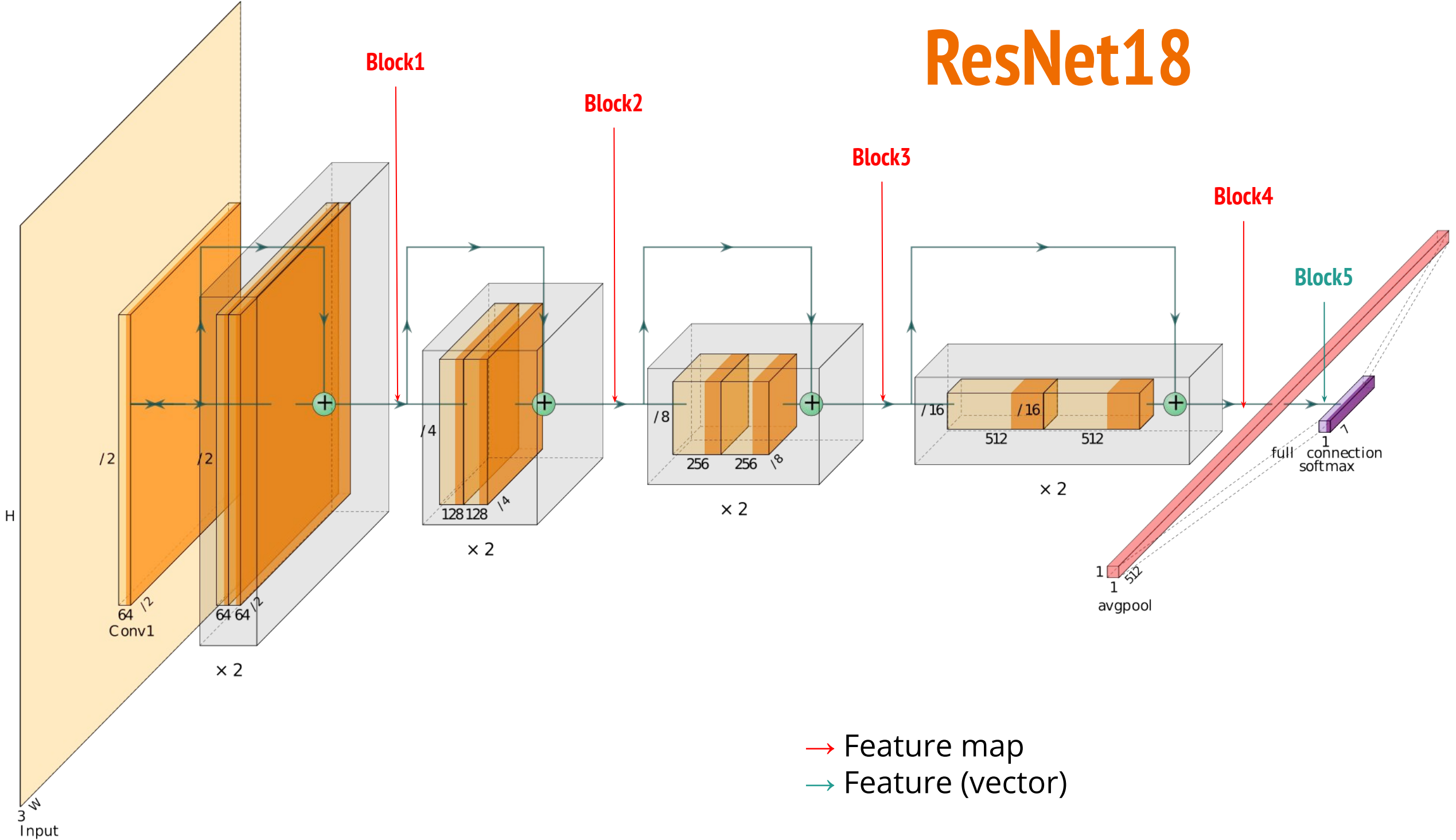
$$\tilde{x}_l = (1 - \mathbf{M}_C) \odot h_l(x_a) + \mathbf{M}_C \odot h_l(x_b),$$

$$\mathcal{L}_{\text{total}} = p_a \mathcal{L}_{\text{feata}}(\hat{y}, y_a) + p_b \mathcal{L}_{\text{featb}}(\hat{y}, y_b),$$

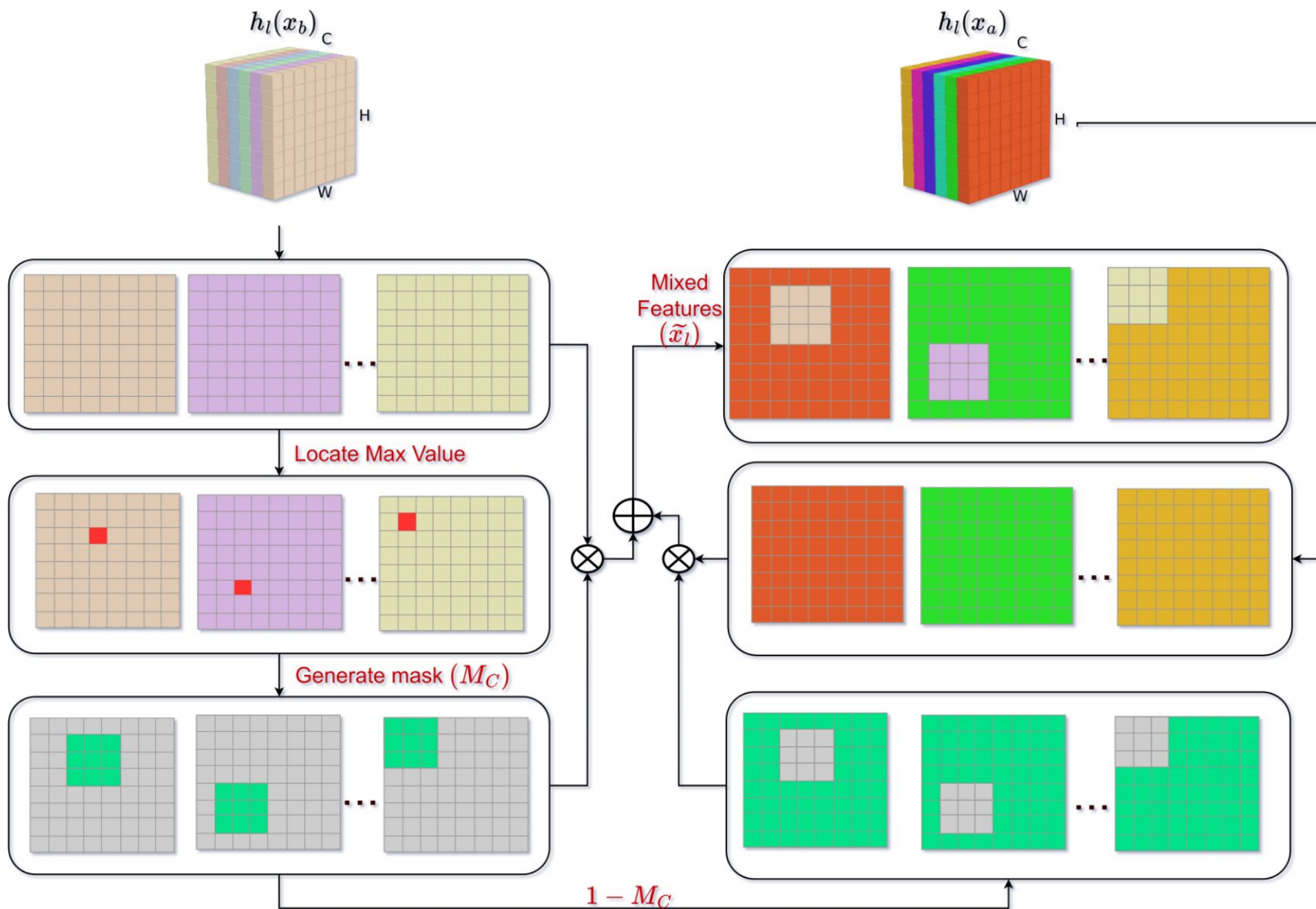
Channels



ResNet18

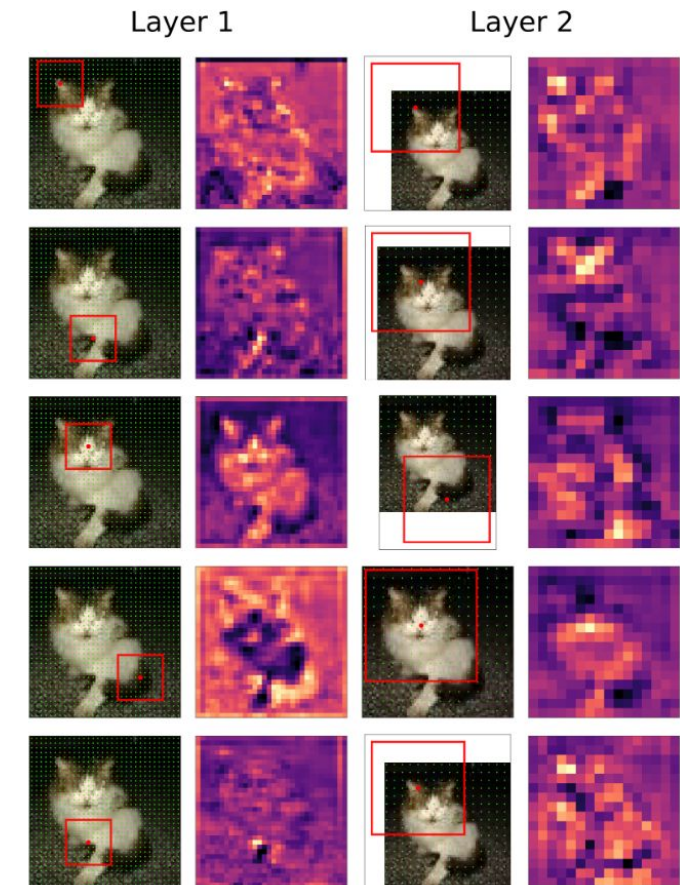


Region-Focused Augmentation Strategy



Channel Max Position Detector

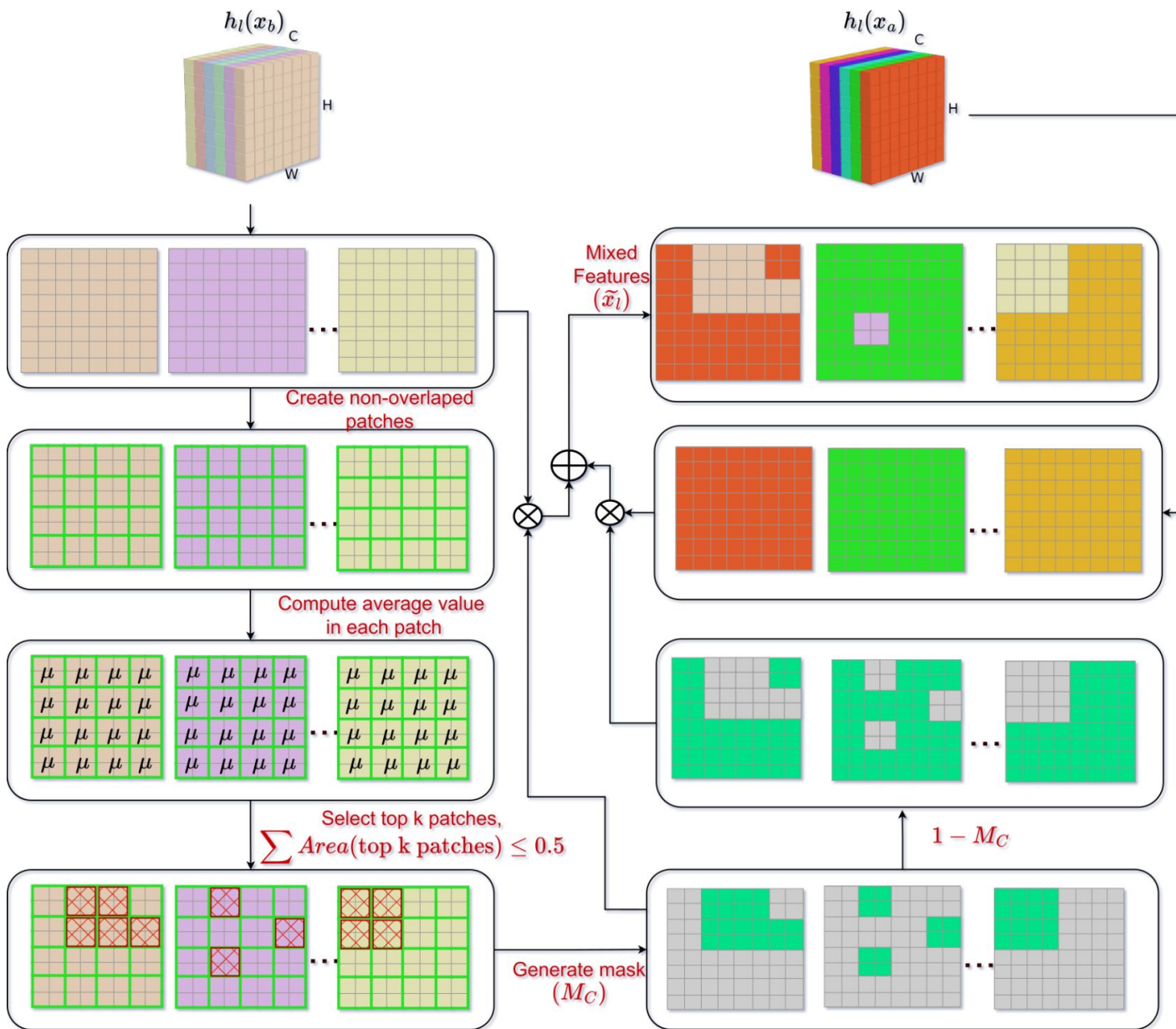
$$\text{idx} = \text{argmax}_c(h_l(x_b)),$$
$$\lambda \sim \text{Uniform}(0, 0.5),$$



Region-Focused Augmentation Strategy

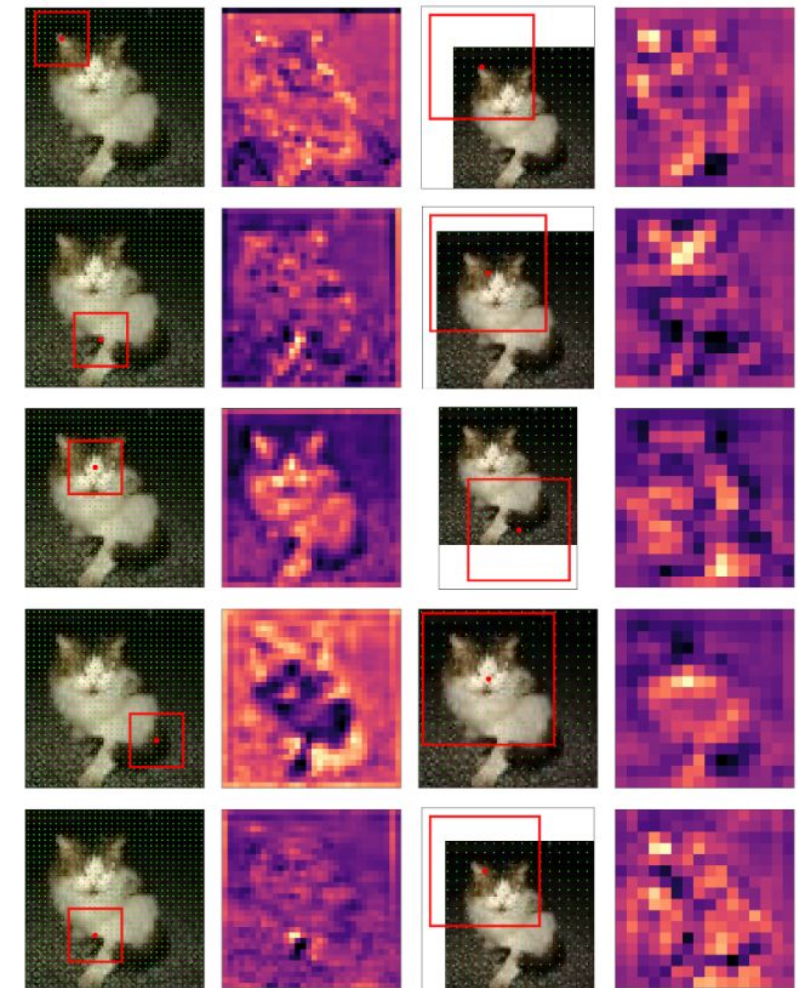
Top k Channel Patches Detector

$$\text{patches}_{\text{nonoverlap}} = S_{\text{splitter}}(h_l(x_b)).$$



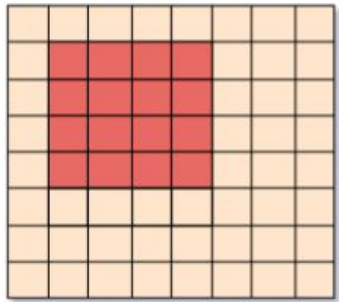
Layer 1

Layer 2



Label Mixing Calibration Method

Area Ratio

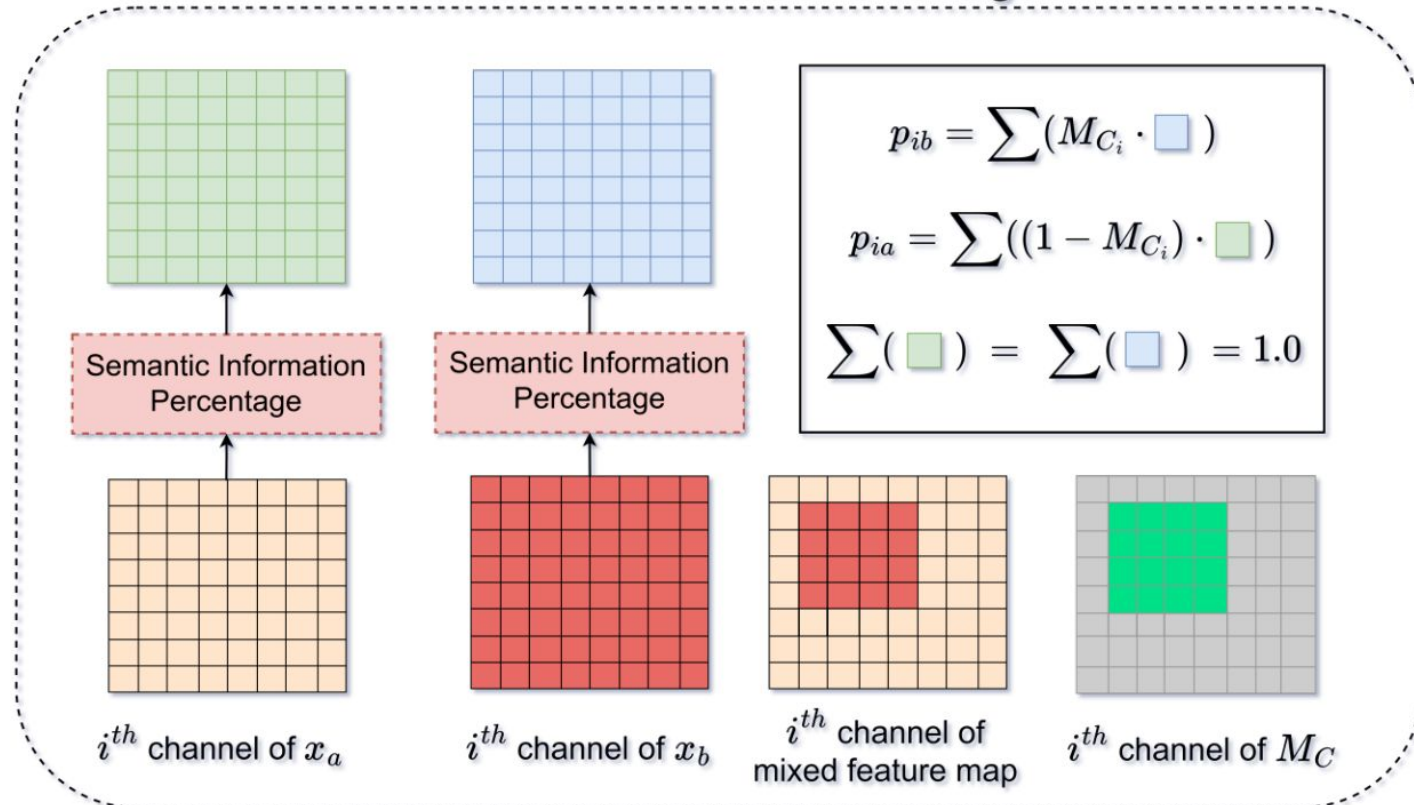


i^{th} channel of mixed feature map

$$p_{ia} = \frac{Area(\text{orange})}{Area(\text{orange}) + Area(\text{red})}$$

$$p_{ib} = \frac{Area(\text{red})}{Area(\text{orange}) + Area(\text{red})}$$

Semantic Information Percentage



FMMix augmentation

TABLE 1. Comparison of four FMMix variants based on our proposed feature map and label mixing methods.

	FMMix1	FMMix2	FMMix3	FMMix4
Max position	✓	✓	✗	✗
Top k patches	✗	✗	✓	✓
Feature map area ratio	✓	✗	✓	✗
Semantic information	✗	✓	✗	✓

TABLE 2. Comparison of Mixup, Manifold Mixup, CutMix, and FMMix. FMMix combines feature-level mixing (similar to Manifold Mixup) and sample-label blending (like Mixup) using a cut-paste approach (from CutMix) with channel-wise and region-focused augmentation strategies.

	Mixup	Manifold Mixup	CutMix	FMMix
Usage of image	✓	✓	✓	✗
Usage of feature map	✗	✓	✗	✓
Channel-Wise	✗	✗	✗	✓
Region-Focused strategy	✗	✗	✗	✓
Regional dropout	✗	✗	✓	✓

Experimental Setup

Experimental Setup

TABLE 5. Summary of the Dataset, Model Complexity, and Basic Configurations for Feature Map Mix Experiments.

Dataset	CIFAR100	Tiny Imagenet	CUB-200-2011	Aircraft	ESC-50
Classes	100	200	200	100	50
Train/Test	50000/10000	100000/10000	5994/5794	6667/3333	1600/400
Model Input Size	32x32	64x64	224x224	224x224	128x250
Pre-trained	No	No	No ImageNet1k	No ImageNet1k	ImageNet1k
Model	ResNet18 ResNet50	ResNet18 ResNet50	EfficientNetV2S	EfficientNetV2S	MobileNetV3Large RexNet100
Batch Size / Epoch	128/200	128/200	64/200	64/200	64/200
Momentum	0.9	0.9	0.9	0.9	0.9
Weight Decay	0.0001	0.0001	0.0001 0.001	0.0001 0.001	0.0001
Initial LR	0.1	0.1	0.01 0.001	0.01 0.001	0.01
LR Decay Factor	0.1	0.1	0.5	0.5	0.5
LR Decay Epoch	100/150	75/150	80/160	80/160	80/160
Model	ResNet18	ResNet50	EfficientNetV2S	MobileNetV3Large	RexNet100
#Params	~12M	~26M	~22M	~5M	~5M
#FLOPs	~2B	~5B	~8.37B	~287M	~510M

Experimental Results

Results

TABLE 1. Comparison of four FMMix variants based on our proposed feature map and label mixing methods.

	FMMix1	FMMix2	FMMix3	FMMix4
Max position	✓	✓	✗	✗
Top k patches	✗	✗	✓	✓
Feature map area ratio	✓	✗	✓	✗
Semantic information	✗	✓	✗	✓

TABLE 6. Accuracy (%) on CIFAR100 employing four FMMix variants across various layers to examine their impact on generalizing the model. (Baseline accuracy: 74.57%.)

Layer	FMMix1	FMMix2	FMMix3	FMMix4
[1]	79.48 (+4.91)	78.85 (+4.28)	78.92 (+4.35)	78.68 (+4.11)
[2]	78.06 (+3.49)	78.45 (+3.88)	78.32 (+3.75)	78.79 (+4.22)
[3]	76.31 (+1.74)	Fail	75.52 (+0.95)	75.90 (+1.33)
[4]	77.39 (+2.82)	76.88 (+2.31)	76.97 (+2.40)	76.94 (+2.37)

TABLE 7. Accuracy (%) on CIFAR100 utilizing four FMMix variants across various combinations of layers to assess their influence on model generalization. (Baseline accuracy: 74.57%.)

Layer	FMMix1	FMMix2	FMMix3	FMMix4
[1, 2]	79.73 (+5.16)	79.51 (+4.94)	79.07 (+4.50)	78.44 (+3.87)
[2, 3]	79.05 (+4.48)	78.93 (+4.36)	79.08 (+4.51)	78.69 (+4.12)
[3, 4]	77.64 (+3.07)	77.52 (+2.95)	76.91 (+2.34)	76.45 (+1.88)
[1, 2, 3]	79.74 (+5.17)	79.47 (+4.90)	80.04 (+5.47)	79.24 (+4.67)
[2, 3, 4]	79.36 (+4.79)	79.66 (+5.09)	79.27 (+4.70)	79.56 (+4.99)
[1, 2, 3, 4]	80.00 (+5.43)	79.88 (+5.31)	79.70 (+5.13)	79.74 (+5.17)

Results

TABLE 8. Accuracy (%) on CIFAR100 and TinyImageNet using Mixup, Manifold Mixup, CutMix, and FMMix. Bold indicates the highest accuracy, and underline indicates the second highest.

Model	Resnet18		Resnet50	
Dataset	Cifar100	Tiny ImageNet	Cifar100	Tiny ImageNet
Baseline	74.57	58.46	74.63	61.52
Mixup	77.48	60.84	79.07	66.52
Manifold Mixup	76.93	54.53	79.96	58.78
CutMix	78.17	62.31	80.42	67.92
FMMix	<u>80</u>	<u>65.19</u>	81.06	<u>69.7</u>
Mixup + FMMix	80.39	65.3	<u>81.38</u>	69.46
CutMix + FMMix	79.74	65.14	81.74	69.87

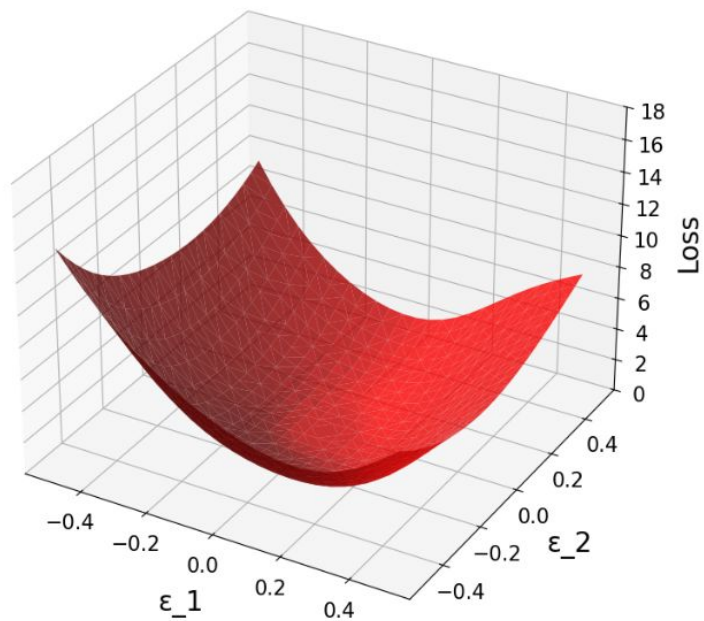
TABLE 9. Accuracy (%) on CUB and Aircraft datasets using EfficientNetV2s, trained both without and with pretraining on ImageNet1K (*). Bold indicates the highest accuracy, and underline indicates the second highest.

Data	CUB	CUB*	Aircraft	Aircraft*
Baseline	43.8	80.74	63.19	78.4
Mixup	<u>50.26</u>	82.98	<u>75.13</u>	81.25
Manifold Mixup	1.69	0.88	2.91	1.83
CutMix	49.36	<u>83.24</u>	68.68	<u>81.94</u>
FMMix	52.64	83.78	75.85	83.2

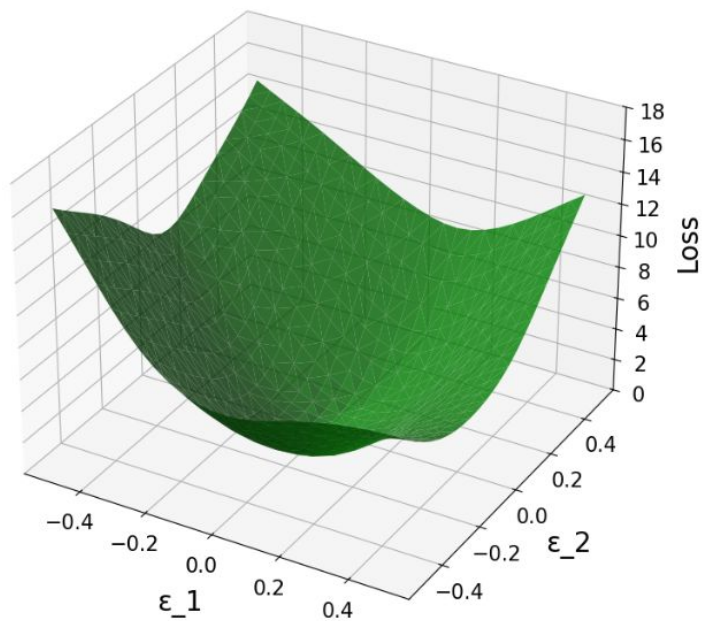
TABLE 10. Accuracy (%) of audio classification task using MobileNetV3Large and RexNet100, both pre-trained on ImageNet1K. Bold indicates the highest accuracy, and underline indicates the second highest.

Model	MobileNetV3Large	RexNet100
Baseline	78.75	80.35
Mixup	82.95	85.7
Manifold Mixup	67.15	55.5
CutMix	<u>87.2</u>	<u>88.5</u>
FMMix	89.35	90.6

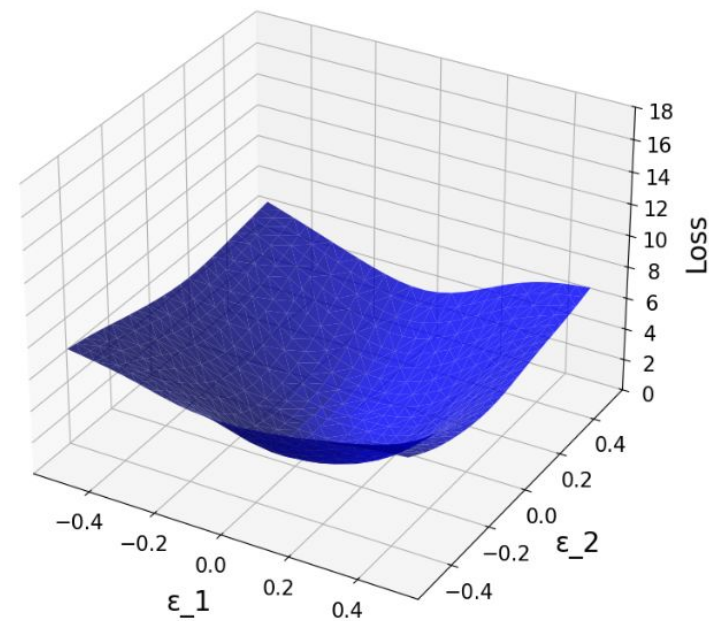
Baseline



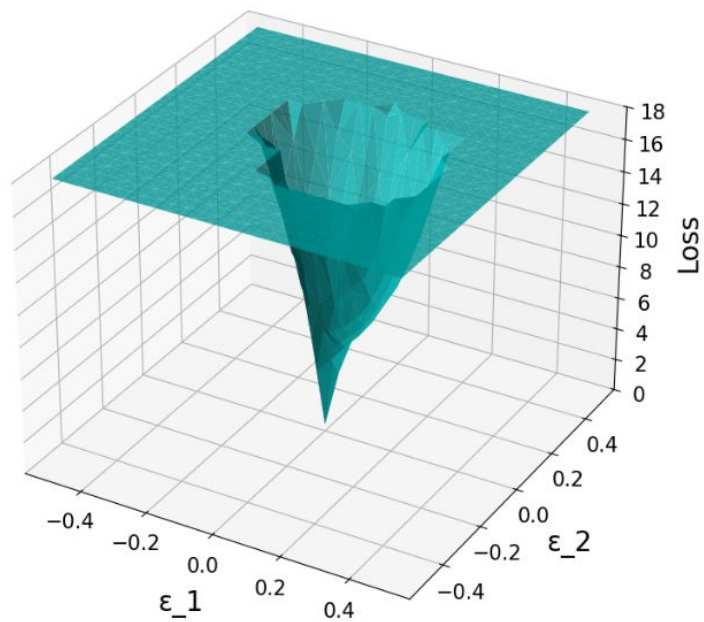
Mixup



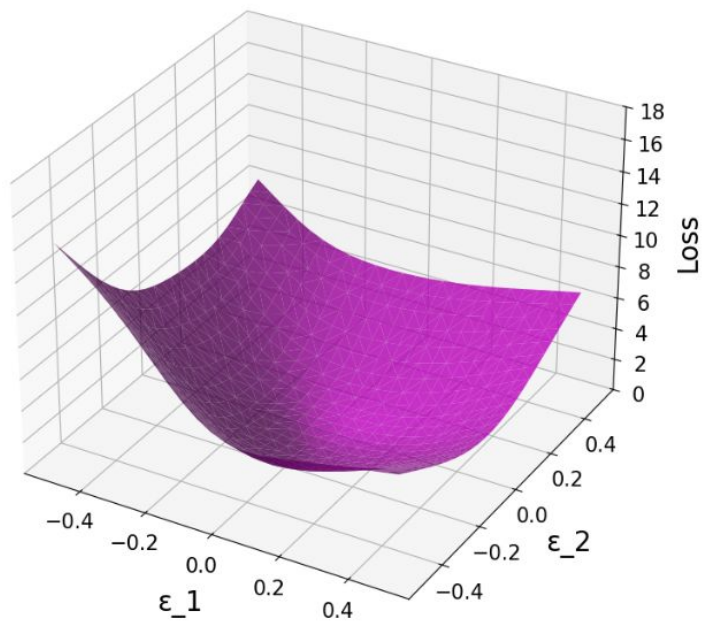
CutMix



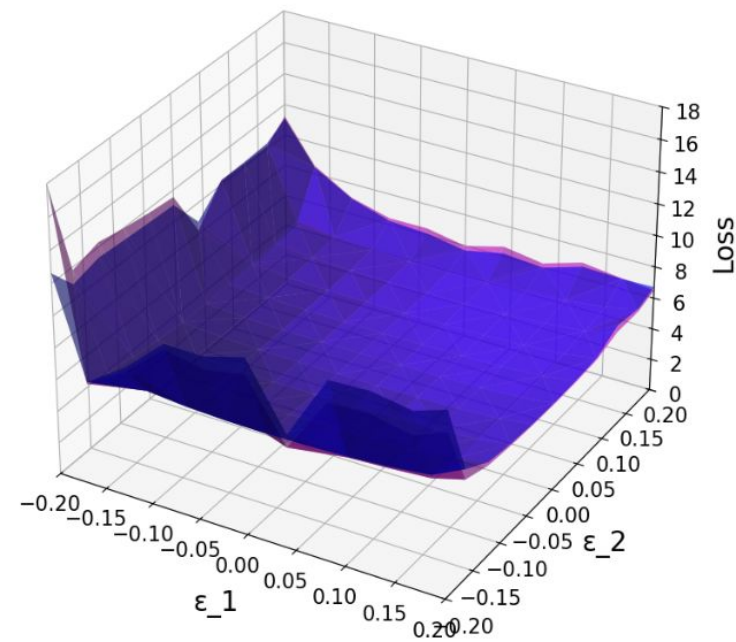
Manifold Mixup



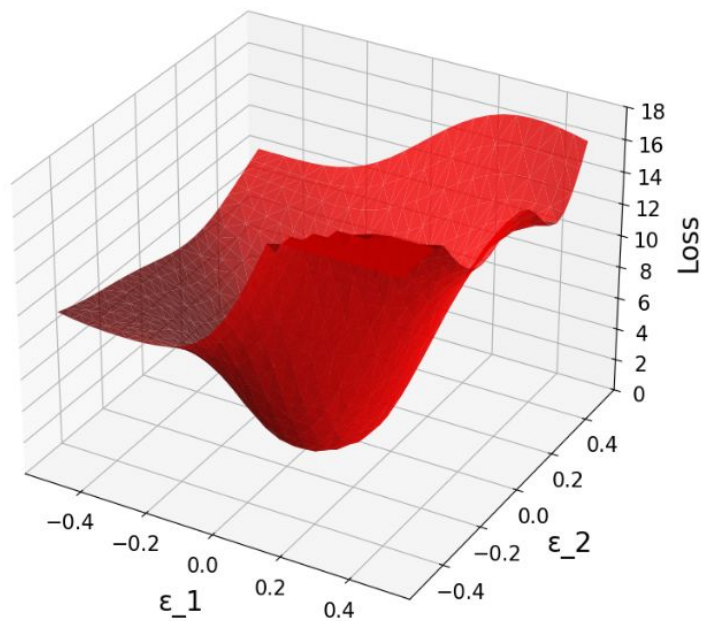
FMMix



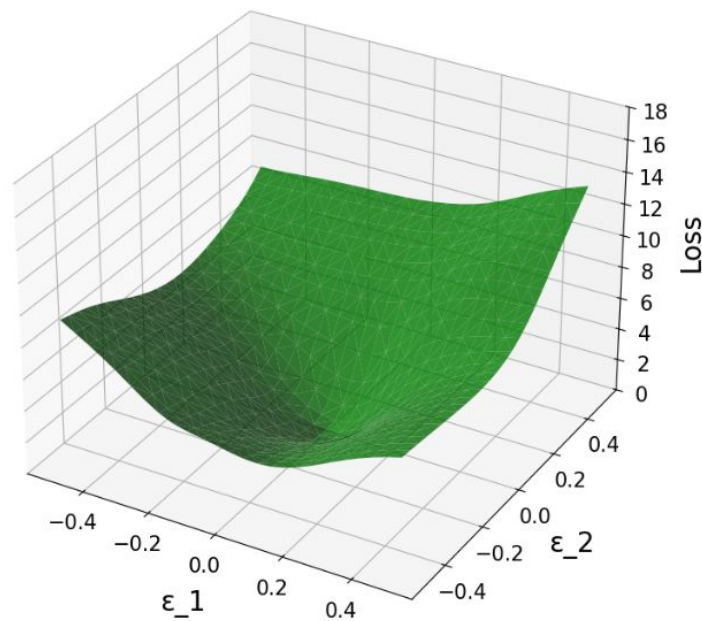
CutMix vs FMMix



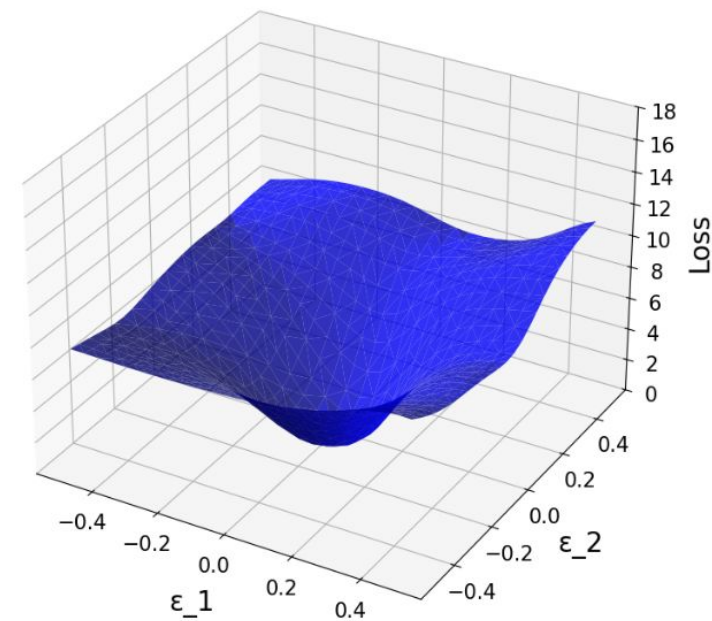
Baseline



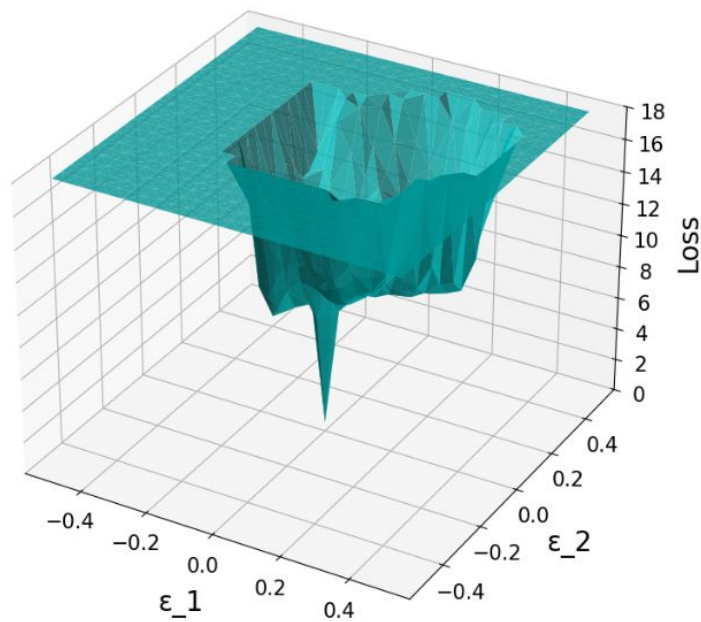
Mixup



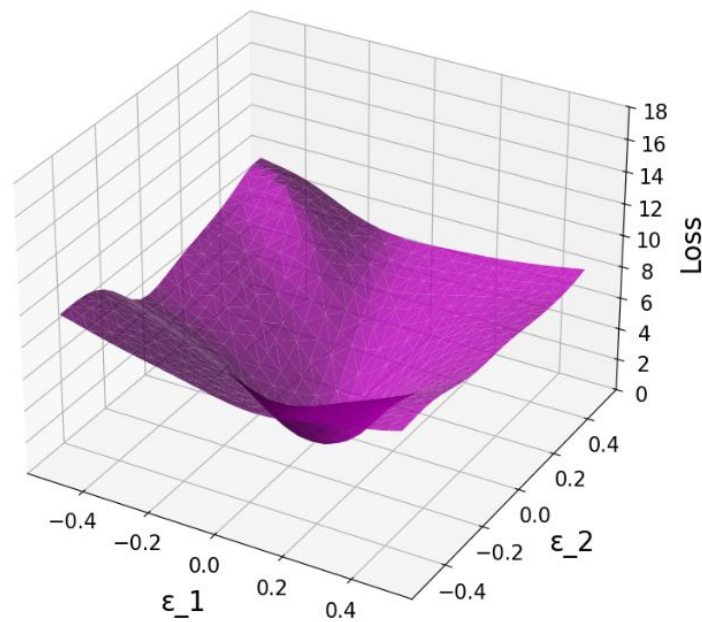
CutMix



Manifold Mixup



FMMix



CutMix vs FMMix

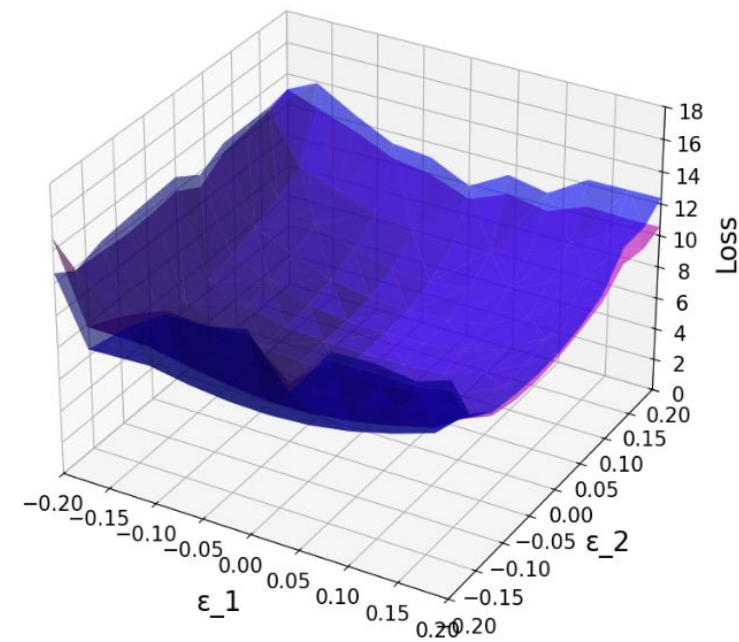


TABLE 11. ECE (10^{-2}) measurements for CIFAR100 and TinyImageNet using ResNet50 and ResNet18. Bold indicates the best results, and underline denotes the second best.

Data/Model	Baseline	Mixup	Manifold Mixup	CutMix	FMMix
CIFAR100-Resnet18	16.09	9.71	19.33	<u>5.03</u>	1.45
CIFAR100-Resnet50	14.72	7.19	10.95	<u>8.87</u>	<u>7.48</u>
TinyImageNet-Resnet18	21.69	11.02	7.29	<u>6.40</u>	3.90
TinyImageNet-Resnet50	17.56	8.11	6.91	<u>9.65</u>	<u>7.87</u>

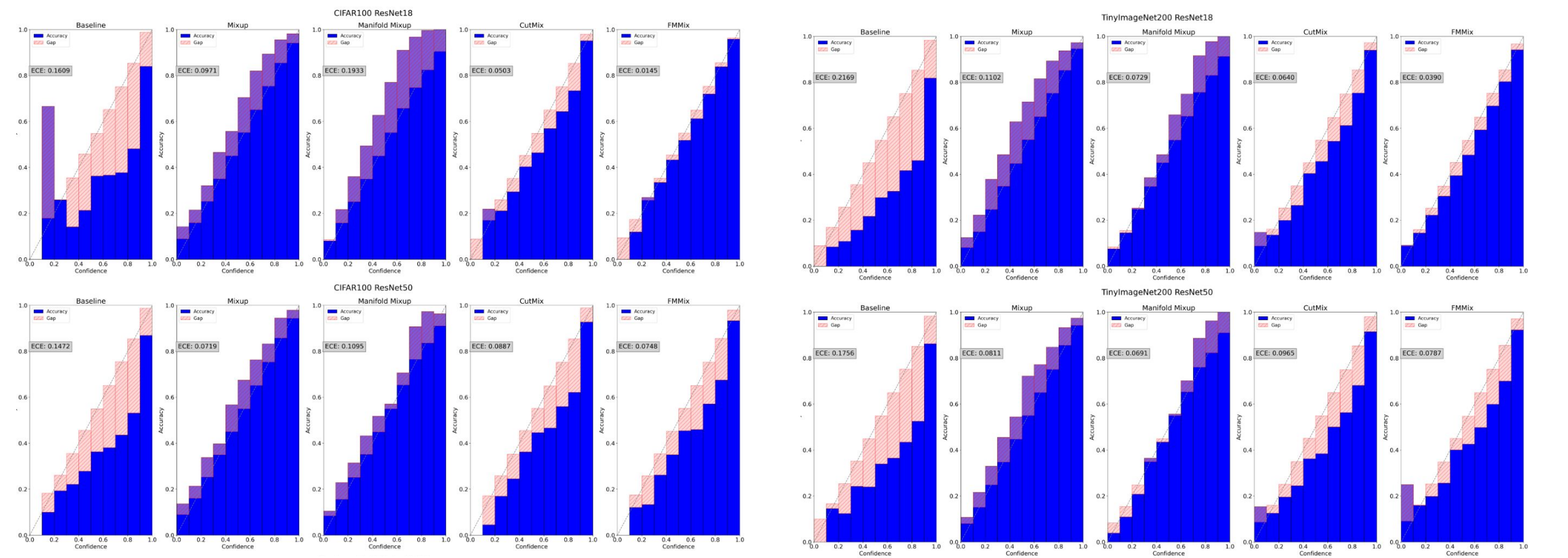


TABLE 12. Measurement of time for one training step and memory for one-step forward processing across datasets and models. The table compares performance for input models from 32x32 to 224x224 across general and fine-grained classification tasks. The highest time and memory consumption is highlighted in bold, while the second highest is underlined.

Data	Model	Type	Time Average (Second)	CPU Average (Byte)	GPU Average (GB)
CIFAR100	ResNet50	Baseline	0.131325	0	8.3764
CIFAR100	ResNet50	Mixup	0.134715	0	8.3764
CIFAR100	ResNet50	Manifold Mixup	<u>0.136725</u>	0	<u>8.3788</u>
CIFAR100	ResNet50	Cutmix	<u>0.133606</u>	0	<u>8.3764</u>
CIFAR100	ResNet50	FMMix	0.140302	3072	8.4651
CUB2011	EfficientNetV2S	Baseline	0.475068	0	27.5911
CUB2011	EfficientNetV2S	Mixup	0.476576	0	27.5911
CUB2011	EfficientNetV2S	Manifold Mixup	0.506806	0	<u>27.618</u>
CUB2011	EfficientNetV2S	Cutmix	0.474641	0	27.5911
CUB2011	EfficientNetV2S	FMMix	<u>0.480483</u>	1536	27.6308

TABLE 15. Impact of removing individual FMMix components. The study evaluates three modified models: one without Mixing Label (ML), one without Channel Wise (CW), and one without Region Focus (RF), compared against the full FMMix and baseline. Bold indicates the highest accuracy, underline denotes the worst performance among the variants (excluding the baseline), and ↓ signifies performance worse than the baseline.

Model	RF	CW	ML	ResNet18		ResNet50		EfficientNetV2				
Dataset				CIFAR100	Tiny ImageNet	CIFAR100	Tiny ImageNet	Aircraft	Aircraft*	Cub2001	Cub2001*	Average
+RF+Cw+ML	✓	✓	✓	80	65.19	81.06	69.7	75.85	83.2	52.64	83.78	7.0138
Cw + ML	✗	✓	✓	79.98	64.92	81.28	67.91	73.15	83.44	52.43	83.34	6.3925
+RF+ML	✓	✗	✓	78.49	63.23	80.25	67.16	71.35	82.54	38.47 ↓	81.01	3.3988
+RF+Cw	✓	✓	✗	77.38	59.94	77.02	62.47	56.11 ↓	81.61	40.11 ↓	82.33	0.2075
Baseline	✗	✗	✗	74.57	58.46	74.63	61.52	63.19	78.4	43.8	80.74	0

TABLE 13. Impact of mask area ratio and mask shape in FMMix augmentation on CIFAR100 and TinyImageNet using ResNet18.

Data	Shape	25%	50%	75%
Cifar100	Square	78.99	80	79.67
Cifar100	Rectangle	78.34	78.73	79.22
TinyImageNet	Square	62.79	65.19	65.58
TinyImageNet	Rectangle	61.51	62.64	63.9

TABLE 14. Impact of FMMix rates on CIFAR100 and TinyImageNet using ResNet18. This study tests FMMix rates ranging from 0% to 100% in 25% increments and includes a dynamic augmentation scheduler that adjusts the FMMix rate every 10 epochs.

FMMix Strategy	CIFAR100	TinyImageNet
0%	74.57	58.46
25%	76.79	61.47
50%	78.25	62.75
75%	79.31	64.03
100%	80	65.19
step_increasing	77.89	62.27
step_decreasing	78.94	64.01

Conclusion

FMMix augmentation

TABLE 1. Comparison of four FMMix variants based on our proposed feature map and label mixing methods.

	FMMix1	FMMix2	FMMix3	FMMix4
Max position	✓	✓	✗	✗
Top k patches	✗	✗	✓	✓
Feature map area ratio	✓	✗	✓	✗
Semantic information	✗	✓	✗	✓

TABLE 2. Comparison of Mixup, Manifold Mixup, CutMix, and FMMix. FMMix combines feature-level mixing (similar to Manifold Mixup) and sample-label blending (like Mixup) using a cut-paste approach (from CutMix) with channel-wise and region-focused augmentation strategies.

	Mixup	Manifold Mixup	CutMix	FMMix
Usage of image	✓	✓	✓	✗
Usage of feature map	✗	✓	✗	✓
Channel-Wise	✗	✗	✗	✓
Region-Focused strategy	✗	✗	✗	✓
Regional dropout	✗	✗	✓	✓

FMMix: A Novel Augmentation Strategy for Enhancing Feature Diversity to Boost Model Generalization for Image Data

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